Where the Hierarchy Breaks: Failure Motifs and Calibrated Self-Diagnosis in HIP-NN

Anonymous Author(s)
Affiliation
Address
email

Abstract

The Hierarchically Interacting Particle Neural Network (HIP-NN) writes a molecular energy as a sum over atoms and hierarchical orders and comes with a built-in reliability signal, the non-hierarchicality R. We ask where this hierarchy breaks, whether its breakdown marks wrong predictions, and whether a calibrated error estimate can be built from it. A hippynn reimplementation that matches the published accuracy at 50,000 training molecules and is within 2% of it at 110,426, trained on QM9 molecules with at most eight heavy atoms reaches an MAE of 0.764 ± 0.008 kcal mol^{-1} on 109,024 nine-heavy-atom molecules, where $17.6 \pm 0.2\%$ of predictions miss chemical accuracy. Screening 68 pre-specified structural, electronic, heteroatom, conformational and novelty motifs on half of this chemistry and confirming them on a stoichiometry-disjoint half with family-wise error control yields 25 hierarchy-breaking motifs, significant in every seed: polycyclic topologies, small-ring strain, ring systems absent from training and zwitterions (error ratios 1.24–10.54). At their atoms the second-order term grows up to 6.3-fold and stops decaying, and the model over-binds the strained frameworks; without the hierarchical regulariser R is uninformative. A four-parameter, single-pass hierarchy-convergence uncertainty is calibrated on held-out chemistry ($\mathrm{ECE}_{\mathrm{rel}} = 0.042 \pm 0.004$) and ranks errors better than R, though a post hoc feature-space distance ranks them better still; with motif indicators it recovers 58% of an eight-model ensemble's advantage. Severe failures are compact (2 motifs explain 80% of errors above 3 kcal mol^{-1}), the 1 kcal mol^{-1} tail is not, and on 3,112,861 off-equilibrium ANI-1x conformations errors are dominated by distortion, which R tracks, while the motifs largely do not transfer.

1 Introduction

Neural-network potentials for organic molecules now reach mean absolute errors (MAEs) far below the 1 kcal mol^{-1} "chemical accuracy" threshold on QM9 [1–5], but the mean hides a tail. The Hierarchically Interacting Particle Neural Network (HIP-NN) [5] reaches 0.256 ± 0.003 kcal mol^{-1} (its Table I), yet $2.26 \pm 0.07\%$ of its test molecules still miss chemical accuracy (Table II) — and these are molecules drawn from the training distribution. In screening, molecular dynamics or active learning a model is routinely asked about molecules unlike its training set; one that cannot tell when it is wrong must be checked by hand.

HIP-NN offers an unusual handle on this problem. It writes the energy as a sum over atoms *and* hierarchical orders, each adding many-body information, and is trained to keep the higher orders small. Its authors observed that a non-hierarchicality R, large when the higher orders fail to decay, correlates with the error (their Fig. 3), and proposed it as a built-in reliability measure. That observation was made in distribution and not taken further: we do not know which structures make the hierarchy

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

fail, whether they are where the energy is wrong, whether an estimate built from the hierarchy is calibrated rather than merely correlated, whether failures concentrate in a few interpretable motifs or spread thinly over chemistry, and whether conclusions drawn on QM9's minimised geometries survive on distorted ones.

We study these questions under controlled extrapolation: a reimplementation of HIP-NN with the paper's architecture, loss and schedule, anchored to the published accuracy, is trained on QM9 molecules with at most eight heavy atoms and asked to predict those with nine. A pre-specified motif grammar is screened, and an uncertainty model fitted, on one stoichiometry-disjoint half of the new molecules and confirmed once on the other; confirmed motifs are traced to the order at which the decomposition stops decaying and tested on off-equilibrium ANI-1x conformations.

We find that (i) the reimplementation reproduces the paper's accuracy to within a few per cent (0.345 ± 0.008 versus 0.354 kcal mol^{-1} at 50,000 molecules), and extrapolating by one heavy atom raises the held-out failure rate from $3.49 \pm 0.22\%$ to $17.6 \pm 0.2\%$; (ii) 25 motifs — polycyclic topologies, small-ring strain, unseen ring systems and electronic motifs led by zwitterions — are confirmed on held-out chemistry under family-wise error control; (iii) at their atoms the second-order term, the first that sees angles and ring closure, grows and decays more slowly, the strained frameworks are over-bound, and without the hierarchical regulariser R carries no information; (iv) a four-parameter hierarchy-convergence uncertainty (HCU) is calibrated on held-out chemistry ($\mathrm{ECE}_{\mathrm{rel}} = 0.042 \pm 0.004$) and, with motif indicators, recovers 58% of the gap to an eight-model ensemble, though a post hoc feature-space distance ranks errors better; severe failures are compact ($K = 2$ at 3 kcal mol^{-1}) and the 1 kcal mol^{-1} tail is not; (v) on off-equilibrium ANI-1x conformations errors are dominated by distortion, which R tracks, and the motifs largely do not transfer.

2 The HIP-NN energy hierarchy as a truncated series

HIP-NN [5] predicts the energy of a configuration $\mathcal{C} = \{(Z_i, \mathbf{r}_i)\}$ as

$$\hat{E}(\mathcal{C}) = \sum_{i=1}^{N_{\mathrm{atom}}} \sum_{n=0}^{N_{\mathrm{int}}} \hat{E}_i^n, \qquad \hat{E}_i^n = \sum_a w_a^n z_{i,a}^{\ell_n} + b^n, \tag{1}$$

where $z_i^{\ell_n}$ are the atomic features after the n-th interaction block. With $N_{\mathrm{int}} = 2$, order 0 is a frozen dressed-atom energy per element; order 1 is a nonlinear function of the radial distribution of neighbour species around atom i (one round of distance-modulated message passing, no angles); order 2 sees the neighbours' environments and hence angles, ring closure and other genuinely many-body information. Training penalises the non-hierarchicality

$$R = \sum_{n=1}^{N_{\mathrm{int}}} \sum_i r_i^n, \qquad r_i^n = \frac{(\hat{E}_i^n)^2}{(\hat{E}_i^n)^2 + (\hat{E}_i^{n-1})^2} \in [0, 1], \tag{2}$$

with weight $\lambda_R = 10^{-2}$. We read Eq. (1) as a truncated series: if the hierarchy works, per-atom terms decay roughly geometrically and the omitted orders $n \geq 3$ are negligible. Two single-pass diagnostics follow: the decay ratio $q_i = |\hat{E}_i^2|/|\hat{E}_i^1|$ (with $r_i^2 = q_i^2/(1 + q_i^2)$), and the position of an atom's hierarchy coordinates $(\hat{E}_i^1, \hat{E}_i^2)$ relative to those of the same element in training, which signals extrapolation even when the decay ratio looks benign.

3 Methods

All choices below, except analyses marked post hoc, were fixed before held-out molecules were examined and written to a hashed pre-registration (`logs/discovery/`) verified by the confirmatory script; changes made on discovery data are listed in `attempts_log.md`.

3.1 Data, partitions and the reimplementation

Data. From the 133,885 QM9 molecules [1] we remove, as the paper does, the 3,054 failing the consistency check and the 11 with difficult optimisations (3 overlap), leaving 130,823. The target is U_0 minus atomic references; its standard deviation (238.0 kcal mol^{-1}) matches the paper's

$\sigma_E \approx 238$ kcal mol^{-1}. The 21,799 molecules with at most eight heavy atoms (C, N, O, F) form the training pool (per seed 1,000 for early stopping, 20,799 for training); the 109,024 molecules with exactly nine form the test set. The code asserts and logs this partition. We train 8 seeds, as the paper's $N_{\text{model}} = 8$. Everything learned from test errors — atlas membership, the uncertainty model, its calibration map and thresholds — is learned on a *discovery* partition, and every held-out number is computed once on the complementary *evaluation* partition. The split is by stoichiometry: the 232 formulas are shuffled with a fixed seed and assigned to discovery until it holds 40% of the molecules (44,484 molecules, 125 formulas), leaving 64,540 (107 formulas) with no isomer seen in discovery.

Reimplementation and anchors. Models are built with hippynn [6], whose network and hierarchical-energy modules are used unchanged (`codebase_notes.md`). The architecture, loss and schedule are the paper's (Appendix Table 3, generated from the run logs): two interaction blocks with three on-site layers each, 80 features, softplus ResNet updates, 20 inverse-distance sensitivities between 1.7 and 10 Bohr with a 15 Bohr cutoff (234,327 parameters; paper: 234k); loss $(\text{MAE}+\text{RMSE})/\sigma_E + \lambda_{L2}(\|w\|^2/\sigma_E^2 + \|W\|^2 + \|V\|^2 + \|\tilde{W}\|^2 + \|\tilde{M}\|^2) + \lambda_R\langle R\rangle$ with $\lambda_{L2} = 10^{-6}$, $\lambda_R = 10^{-2}$, implemented by training on E/σ_E (the paper's $w = \sigma_E\tilde{w}$); frozen least-squares order-0 energies and order-n read-outs scaled by 10^{-2n}; Adam (10^{-3}; 0.9, 0.999), minibatch 30, learning rate held for 100 epochs and then halved after 50 epochs without validation improvement, stop after two halvings without improvement or 2000 epochs, best-validation parameters restored; energy-only. Disclosed differences: hippynn's default (Xavier-normal) initialisation instead of Glorot-uniform with zero biases (tested in a diagnostic), and float32 on H200 GPUs. Seeds 0–2 are retrained with $\lambda_R = 0$ as an ablation (the frozen analyses of Secs. 3.2–3.3 were re-applied to them post hoc). To anchor accuracy we repeat the paper's Table I protocol on random splits of all of QM9 with $N_{\text{train}} + N_{\text{valid}} = 110{,}426$ (4 seeds, fewer than the paper's eight for cost), 50,000 (3) and our pool size 21,799 (3).

3.2 Motif grammar, atlas selection and confirmatory tests

Before any error was inspected we wrote 68 candidate motifs, each a rule evaluable on a new molecule (Appendix Table 5): 15 *structural* (ring sizes; fused, bridged, spiro and caged systems; sp^2 ring-fusion atoms), 23 *electronic* (aromaticity, conjugation, triple bonds, C=N, carbonyl families, charge separation), 16 *heteroatom* (N–N, N–O bonds, anomeric carbons, fluorination, heteroatom-rich compositions), 8 *conformational*, evaluated on the B3LYP geometry (close contacts, stretched bonds, acute sp^2 angles, non-planar amide nitrogens, twisted conjugation), and 6 *novelty* rules (a Morgan environment [7], ring system or Bemis–Murcko scaffold [8] absent from the training pool). Graphs are built with RDKit [9] by mapping each GDB SMILES onto its geometry by subgraph isomorphism; each rule also returns the atoms carrying the motif.

On the discovery molecules we compute, per seed, the ratio of the MAE of matching to non-matching molecules and the same ratio for R/N_{atom}. A candidate enters the hierarchy-breaking motif atlas (HBM atlas) if it matches at least 200 discovery molecules but at most half of them and its seed-averaged error and R/N_{atom} ratios are at least 1.25 and 1.10. This is post-hoc identification on test residuals; its validity rests on sample splitting. On the evaluation molecules each atlas motif receives its coverage, conditional error distribution and two one-sided Mann–Whitney tests [10]: larger seed-averaged absolute errors than the other molecules, and a larger order-2 ratio term r_i^2 on motif atoms than on other atoms of the same element. The family of $2 \times |\text{atlas}|$ tests is controlled with Holm's procedure [11] at $\alpha = 0.05$; selection used only discovery data, so no further selective-inference correction is needed. Sensitivity analyses repeat the error test for every candidate with at least 20 matches (Holm over that family) and per seed.

A failure is $|\hat{E} - E| > 1$ kcal mol^{-1} (chemical accuracy, the convention of the paper's Table II) in every precision, recall and compactness statement; compactness is additionally reported at 2 and 3 kcal mol^{-1} as a disclosed secondary analysis. Motifs are ordered by their failure attribution on discovery data (a greedy set-cover order is a secondary view); per seed, the cumulative attribution of the top k motifs is the fraction of failures in the union of their matches, K is the smallest k reaching 80%, and the fraction of all molecules in the same union is reported because broad motifs reach high attribution trivially. Mechanistic contrasts compare r_i^1, r_i^2, $\log|\hat{E}_i^n|$, $\log q_i$ and A_i (below) on motif atoms with other atoms of the same element (seed-averaged).

Table 1: Two-baseline summary. Paper: Table I of Ref. [5] and a power-law extrapolation of its points to 21,799. Reimplementation: mean ± s.e. over seeds; the last two rows are the same eight size-extrapolating models on all 109,024 nine-heavy-atom and on the 64,540 held-out evaluation molecules. $\mathrm{ECE}_{\mathrm{rel}}$ of HCU; K_{80}: atlas motifs covering 80% of failures ($|\hat{E} - E| > 1$ kcal mol^{-1}; median over seeds).

Model	Split	$N_{\mathrm{tr}}+N_{\mathrm{val}}$	Test set	MAE (kcal mol^{-1})	$\mathrm{ECE}_{\mathrm{rel}}$	K_{80}
Paper (Table I)	random	110,426	remainder	0.256 ± 0.003	–	–
Paper (Table I)	random	50,000	remainder	0.354 ± 0.004	–	–
Paper (extrapolated)	random	21,799	remainder	0.501	–	–
Reimplementation (4 seeds)	random	110,426	remainder	0.262 ± 0.007	–	–
Reimplementation (3 seeds)	random	50,000	remainder	0.345 ± 0.008	–	–
Reimplementation (3 seeds)	random	21,799	remainder	0.545 ± 0.008	–	–
Reimplementation (8 seeds)	≤8 heavy	21,799	all 9-heavy	0.764 ± 0.008	–	–
This work (HCU + atlas)	≤8 heavy	21,799	held-out 9-heavy	0.732 ± 0.007	0.042 ± 0.004	10

3.3 Hierarchy-convergence uncertainty (HCU)

HCU turns one forward pass into a calibrated predictive distribution for $|\varepsilon| = |\hat{E} - E|$ from four signals: (1) the paper's R; (2) a geometric tail estimate of the omitted orders, $T = \sum_i |\hat{E}_i^2|\, \tilde{q}_i/(1-\tilde{q}_i)$ with $\tilde{q}_i = \min(q_i, 0.95)$ — the remainder $\sum_{n\geq 3} \hat{E}_i^n$ if each atom's series kept decaying at its observed ratio, a standard truncation-error estimate that, unlike R, carries an energy scale; (3) the mean atypicality $\bar{A}$ of the atoms' hierarchy coordinates, where A_i is the distance from atom i's standardised $(\hat{E}_i^1, \hat{E}_i^2)$ to its tenth nearest neighbour among training atoms of the same element, divided by the 99th percentile of that distance within training; (4) N_{atom}, since errors are extensive. We fit $\log \mu_{\mathrm{raw}} = \beta_0 + \beta_1 \log R + \beta_2 \log T + \beta_3 \log \bar{A} + \beta_4 \log N_{\mathrm{atom}}$ with $|\varepsilon| \sim \mathrm{Exponential}(\mu)$ by maximum likelihood on discovery molecules, per seed, recalibrate isotonically [12, 13], and read $P(|\varepsilon| > \tau)$ from the empirical distribution of $|\varepsilon|/\mu$ on discovery molecules. Comparators: R alone (same isotonic map), HCU plus atlas indicators, and the eight-seed ensemble spread [14–16] alone and with HCU. On evaluation molecules, predictions are binned into ten equal-mass bins of μ; the calibration curve shows all bins, and the primary scalar is $\mathrm{ECE}_{\mathrm{rel}} = \sum_b \frac{n_b}{N} |\bar{\varepsilon}_b - \bar{\mu}_b|/\bar{\mu}_b$, the mean-error analogue of the ENCE [17]; we also report the largest bin deviation, the classification ECE [18, 19] of $P(|\varepsilon| > 1 \text{ kcal mol}^{-1})$, coverage of error bounds, Spearman correlation and AUROC. Failures are flagged at two thresholds fixed on discovery data: $P \geq 0.5$ (Bayes-optimal for equal costs if calibrated) and the F_1-optimal threshold. Post hoc we added a latent-distance comparator [20]: the same atypicality in the 80-dimensional final-layer feature space (mean $\bar{L}$), alone, with N_{atom}, and added to HCU.

3.4 Conformational transfer

We take *all* ANI-1x [21] (ωB97X/6-31G(d)) conformations of molecules with at most nine heavy atoms (4,194,049 of 4,956,005). Molecules are identified within formula groups by their bonded graph, and the grammar is evaluated on each molecule's lowest-energy conformation c_0; charged, fragmented or unperceived molecules are excluded. To remove all molecule-level offsets between B3LYP U_0 and ωB97X energies, errors are those of relative conformational energies, $\tilde{\varepsilon}_c = (\hat{E}_c - \hat{E}_{c_0}) - (E_c - E_{c_0})$, with distortion energy $\Delta E_c = E_c - E_{c_0}$. The unit of inference is the molecule, and its primary outcome adjusts for distortion: the mean of $\log|\tilde{\varepsilon}_c|$ minus the global binned-median curve of $\log|\tilde{\varepsilon}|$ versus $\log \Delta E$. Atlas motifs are tested with permutation tests stratified by heavy-atom count (4,000 permutations; Holm over the atlas); the transfer ratio is $\log \mathrm{ER}_{\mathrm{ANI}}/\log \mathrm{ER}_{\mathrm{QM9}}$ of geometric-mean error ratios, and a motif transfers if its enrichment is positive and Holm-significant. Frozen HCU models are applied unchanged; all conformations of the ten rMD17 molecules [22] provide a descriptive check (Appendix).

4 Results

We first anchor the baseline and measure what extrapolation costs, then confirm the atlas and trace its mechanism, and finally evaluate self-diagnosis, compactness and transfer.

4.1 A faithful baseline, and what size extrapolation costs it

On random splits of all of QM9 (the paper's protocol) the reimplementation reaches 0.345 ± 0.008 kcal mol^{-1} at 50,000 molecules, at the published accuracy (0.354 ± 0.004), and 0.262 ± 0.007 kcal mol^{-1} at 110,426, 2% above the published 0.256 ± 0.003 with the same tail ($2.37 \pm 0.15\%$ of errors above 1 kcal mol^{-1}; paper $2.26 \pm 0.07\%$). At our pool size it gives 0.545 ± 0.008 kcal mol^{-1}, 9% above a power-law extrapolation of the paper's points (0.501). Training with the paper's initialisation, our one procedural difference, shifts these MAEs by only 0.004 ± 0.020 and 0.008 ± 0.007 kcal mol^{-1} (21,799 and 50,000 molecules; Appendix B). Where the paper can be checked directly the baseline matches it to within 2%, slightly weaker than published at the largest size.

The eight size-extrapolating models (1556 ± 100 epochs, all stopped by the paper's termination rule) reach 0.312 ± 0.017 kcal mol^{-1} on in-distribution validation molecules, with $3.49 \pm 0.22\%$ of errors above 1 kcal mol^{-1} (paper: $2.26 \pm 0.07\%$). On nine-heavy-atom molecules the MAE rises to 0.764 ± 0.008 kcal mol^{-1} and $17.6 \pm 0.2\%$ of predictions miss chemical accuracy. At matched data, a random-split model that saw nine-heavy-atom molecules reaches 0.565 ± 0.008 kcal mol^{-1} (13.1% failures) on the held-out molecules in its test set against 0.731 ± 0.007 kcal mol^{-1} (17.1%): extrapolating by one heavy atom costs 29% in MAE.

On average the decomposition remains a hierarchy (truncated after orders 0, 1, 2 the held-out MAE is 20.62, 2.74, 0.73 kcal mol^{-1}; median decay ratio q_i 0.023 on validation and 0.027 on evaluation atoms), and the paper's uncertainty claim survives extrapolation: error quantiles rise monotonically with R both in distribution (Spearman 0.29) and on held-out molecules (0.32; Appendix Fig. 4). The hierarchical regulariser creates this signal: with $\lambda_R = 0$ the MAE barely changes (0.755 ± 0.016 versus 0.736 ± 0.016 kcal mol^{-1}, seeds 0–2) but R grows 19-fold and loses its information about the error (Spearman 0.044 ± 0.067 versus 0.324 ± 0.019; AUROC 0.540 ± 0.048 versus 0.736 ± 0.011) — unpenalised, the higher orders carry arbitrary shares of the energy. Under the frozen atlas these models still fail on all 25 motifs (error ratios correlate at 0.99 with the main models), but only 13 (versus 25) show an order-2 excess at the motif atoms (Appendix Table 9): the regulariser does not create the failures, it makes the hierarchy localise them.

4.2 The hierarchy-breaking motif atlas

25 of the 68 candidates pass the discovery screen, and on held-out molecules all 25 carry significantly larger errors after Holm correction, each also significant in every seed separately (Table 2, Fig. 1). Error ratios run from 1.24 (aromatic six-rings) to 10.54 (zwitterions), and the tails are what matter: the 99th error percentile is 22.8 kcal mol^{-1} for novel ring systems against 7.26 kcal mol^{-1} overall. The atlas falls into four groups: polycyclic topology, small-ring strain and its geometric signatures, novelty with respect to the training pool (unseen ring systems and scaffolds: error ratios 4.44 and 3.71), and electronic and heteroatom motifs. The order-2 ratio term is elevated on the motif atoms for 24 motifs (for H15 the rank test is significant but the mean shift is nil). Over all 61 testable candidates (Appendix Table 6), 38 show significant excess error and 22 are *easier* than average (acyclic molecules 0.54, alkynes 0.54, nitriles 0.69). The significant candidates outside the atlas are near-threshold electronic motifs or failures that do *not* raise R — rings of seven or more atoms (error ratio 1.84 at a discovery R ratio of 1.01), C=N (1.40) and N–N bonds (1.49) — discussed in Sec. 5.

4.3 Mechanism: the second order stops decaying

The breakdown sits in the second interaction block (Fig. 1c). On motif atoms the order-1 ratio term barely moves ($|\Delta r^1| \leq 8.0 \times 10^{-3}$, order 1 being dwarfed by the dressed-atom energy), whereas $|\hat{E}_i^2|$ is larger for 23 of 25 motifs, by factors of 1.4–6.3, and the decay ratio q_i grows for 17 — most for short H$\cdots$H contacts, zwitterions, amidines, sp^2 ring-fusion atoms and twisted conjugation ($e^{\Delta \log q} = 6.0, 3.6, 2.6, 2.5, 2.2$). For the other eight both orders grow together, which the atypicality A_i registers (higher on motif atoms for 24 motifs). This is what the architecture predicts: order 1 depends only on the radial distribution of neighbour species, so angles, ring closure, conjugation and charge separation enter only through order 2 — exactly the terms in which a nine-atom cage, a pyramidalised sp^2 ring-fusion centre or a zwitterion differs from its eight-atom analogues.

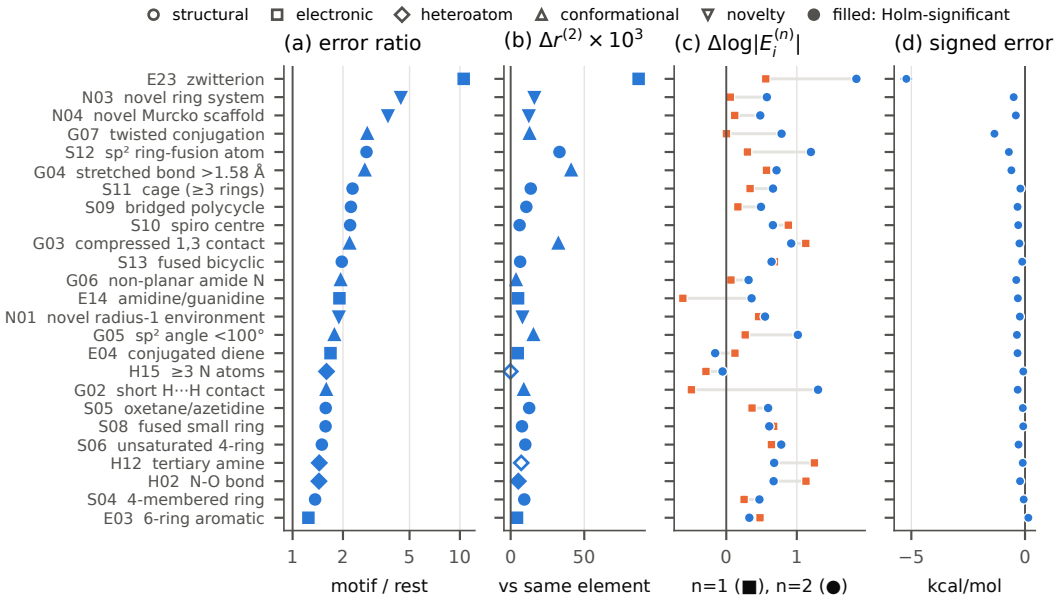


Figure 1: The HBM atlas on held-out molecules, ordered by error ratio (seed-averaged; markers encode the motif family, filled = Holm-significant). (a) Error ratio of motif to remaining molecules with seed s.e.; (b) element-matched excess of the order-2 ratio term r_i^2 on the motif atoms; (c) element-matched shift of $\log|\hat{E}_i^1|$ (squares) and $\log|\hat{E}_i^2|$ (circles); (d) mean signed error (negative: predicted too stable).

The error has a direction (Fig. 1d). 24 of 25 motifs are over-bound, most strongly twisted conjugation (-1.34 kcal mol^{-1}), sp^2 ring-fusion atoms (-0.71), stretched bonds (-0.60) and zwitterions (-5.22): the model assembles an unfamiliar strained framework from familiar pieces and misses the non-additive destabilisation [23]. The extreme tail consists of cages with a pyramidalised Bredt-type C=C or C=N at a ring fusion, predicted up to 59 kcal mol^{-1} too stable. Within a motif the molecular order-2 sum explains little of the error variance ($R^2 \leq 0.09$): the hierarchy says *where* the model extrapolates, not by how much.

Coverage versus intrinsic difficulty (Appendix Table 8). For the size-matched random-split model, which saw nine-heavy-atom molecules, the excess error of novelty and topology motifs shrinks (novel ring systems 4.44→2.71, spiro centres 2.21→1.52) — a coverage effect — while electronic motifs keep it (zwitterions 10.54 versus 10.02, amidines 1.91 versus 2.04), being intrinsically hard for a local, energy-only model [4].

4.4 Calibrated self-diagnosis

All predictors are calibrated in the mean (Fig. 2a; Appendix Table 7). For HCU, $\mathrm{ECE}_{\mathrm{rel}} = 0.042 \pm 0.004$ (worst seed 0.063); its bins deviate from the identity by at most 9.3% on average, within $\pm 10\%$ for 5 of eight seeds (range 5.7–13.1%), and nominal 50/90/95% error bounds cover 49.1/89.6/94.9% of molecules. Since a monotone map calibrates any score (R alone: $\mathrm{ECE}_{\mathrm{rel}} = 0.034 \pm 0.004$), discrimination is the informative comparison. HCU ranks errors better than R (Spearman 0.367 ± 0.008 versus 0.331 ± 0.009; AUROC 0.770 ± 0.005 versus 0.745 ± 0.006), closing 22% of the AUROC gap to the eight-model ensemble (0.857 ± 0.002); with atlas indicators it closes 58% (0.810 ± 0.003, $\mathrm{ECE}_{\mathrm{rel}} = 0.028 \pm 0.003$). Adding HCU to the ensemble changes nothing (0.858 ± 0.002). At the rule $P(|\varepsilon| > 1 \text{ kcal mol}^{-1}) \geq 0.5$, HCU flags 6.6% of molecules with precision 0.61 and recall 0.24 (false-positive rate 0.03, false-negative rate 0.76; base rate 17.1%), a conservative alarm suited to expensive recomputation; the F_1-optimal threshold, suited to screening triage, flags 23.1% with precision 0.40 and recall 0.54 (with atlas indicators 0.49 and 0.54). Keeping the half of molecules with the lowest predicted error lowers the MAE from 0.732 ± 0.007 to 0.397 kcal mol^{-1}.

The post hoc latent-distance comparator qualifies this result: the same atypicality measured in the 80-dimensional feature space ranks errors better than HCU in 8 of eight seeds (AUROC 0.802 ± 0.004 with N_{atom}; $\mathrm{ECE}_{\mathrm{rel}} = 0.035 \pm 0.003$), closing 50% of the gap to the ensemble, and adding the

Table 2: HBM atlas statistics on the 64,539 held-out molecules (means over eight seeds). Fam.: S structural, E electronic, H heteroatom, C conformational, N novelty. Cov.: % of molecules carrying the motif; MAE, Q_{90}, Q_{99}: mean and percentiles of $|\hat{E} - E|$ (kcal mol^{-1}); Fail: % with $|\hat{E} - E| > 1$ kcal mol^{-1}; ER: MAE ratio to the other molecules; p_{Holm}: one-sided Mann–Whitney test of seed-averaged errors, Holm-adjusted in the family of 50 tests; $\Delta r^{(2)}$: element-matched excess of the order-2 ratio term on motif atoms; ER$_{\text{ANI}}$: distortion-adjusted ratio on ANI-1x (†: Holm-significant transfer).

ID	Motif	Fam.	Cov. %	MAE	Q_{90}	Q_{99}	Fail %	ER	p_{Holm}	$\Delta r^{(2)} \times 10^3$	ER$_{\text{ANI}}$
S13	fused bicyclic	S	40.0	1.04	2.01	10.4	25	1.97	$<10^{-300}$	+6	0.99
S08	fused small ring	S	46.2	0.91	1.68	9.4	21	1.57	$<10^{-300}$	+8	1.02†
S04	4-membered ring	S	39.9	0.87	1.57	9.2	19	1.36	10^{-193}	+9	1.01
S11	cage (≥3 rings)	S	24.6	1.27	2.47	13.4	30	2.28	$<10^{-300}$	+14	1.05†
N04	novel Murcko scaffold	N	12.3	2.03	4.31	19.7	47	3.71	$<10^{-300}$	+12	0.98
N03	novel ring system	N	8.9	2.48	5.56	22.8	54	4.44	$<10^{-300}$	+16	1.00
S05	oxetane/azetidine	S	21.1	1.03	1.88	11.7	23	1.58	10^{-272}	+13	1.02†
S09	bridged polycycle	S	13.5	1.40	2.71	15.6	30	2.23	$<10^{-300}$	+11	1.07†
G03	compressed 1,3 contact	C	11.8	1.40	2.98	15.6	30	2.19	$<10^{-300}$	+32	1.02†
N01	novel radius-1 environment	N	11.5	1.25	2.37	15.3	28	1.89	$<10^{-300}$	+8	1.04†
G05	sp^2 angle $<100°$	C	11.5	1.19	2.19	14.1	26	1.78	10^{-200}	+15	1.01
G04	stretched bond >1.58 Å	C	8.0	1.74	3.91	19.9	32	2.70	$<10^{-300}$	+41	1.11†
H15	≥3 N atoms	H	9.1	1.10	2.21	8.4	32	1.60	$<10^{-300}$	0	1.06†
H12	tertiary amine	H	11.6	1.00	1.78	10.5	24	1.44	10^{-181}	+7	1.04†
S12	sp^2 ring-fusion atom	S	4.7	1.87	3.89	19.0	41	2.77	$<10^{-300}$	+33	0.99
S10	spiro centre	S	9.1	1.45	3.36	16.8	26	2.21	10^{-146}	+6	0.96
S06	unsaturated 4-ring	S	8.8	1.05	1.97	12.3	23	1.50	10^{-72}	+10	0.85
E03	6-ring aromatic	E	4.3	0.90	1.94	6.4	27	1.24	10^{-127}	+4	0.88
H02	N-O bond	H	4.3	1.03	2.10	7.5	28	1.44	10^{-130}	+5	1.07†
E14	amidine/guanidine	E	4.0	1.34	2.46	16.7	32	1.91	10^{-222}	+5	1.06†
G06	non-planar amide N	C	3.6	1.37	2.59	16.4	32	1.94	10^{-195}	+4	1.04†
E23	zwitterion	E	0.4	7.42	21.98	39.4	74	10.54	10^{-121}	+87	–
G02	short H$\cdots$H contact	C	1.4	1.15	2.44	8.9	32	1.59	10^{-84}	+9	1.11†
E04	conjugated diene	E	0.2	1.23	2.54	10.6	32	1.68	10^{-10}	+5	0.97
G07	twisted conjugation	C	0.3	2.03	4.66	18.2	48	2.79	10^{-41}	+13	1.01

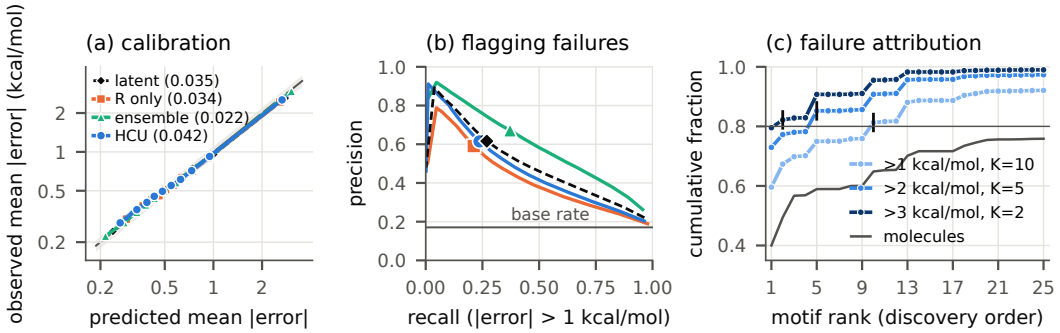


Figure 2: Held-out uncertainty and compactness. (a) Observed against predicted mean $|\hat{E} - E|$ in ten equal-mass bins, ECE$_{\text{rel}}$ in parentheses (thick: mean over seeds; thin: HCU per seed; grey band: ±10% of the identity; dashed: post hoc latent-distance comparator $\bar{L} + N_{\text{atom}}$). (b) Precision–recall of flagging failures ($|\hat{E} - E| > 1$ kcal mol^{-1}) over a threshold sweep; markers: pre-specified $P \geq 0.5$ rule. (c) Cumulative fraction of failures in the union of the top-k atlas motifs (order fixed on discovery data) for three thresholds, and fraction of all molecules covered (grey); ticks and legend mark K, the first rank reaching 80% (motif identities: Appendix Fig. 5).

hierarchy signals changes little (0.807 ± 0.004; Appendix Table 7). The hierarchy coordinates are a physically labelled two-dimensional projection of the representation (Spearman 0.76 between the atypicalities); they add an interpretation — which order fails, on which atoms, in which direction — not a better ranking.

4.5 How compact are the failures?

At chemical accuracy failures are frequent (17.1% of held-out molecules) and only moderately compact (Fig. 2c). In the pre-specified order the top five motifs contain 75.0% of failures while

covering 58.9% of molecules, and $K = 10$ (10–11 across seeds) motifs are needed for 80%, because the leading motifs are nested ring-topology classes. The set-cover order fixed on discovery data (S13, H15, S04, N01, N04) needs $K = 4$, but its top five also cover 67.6% of molecules (failure rate $1.28\times$ the average). Molecules outside the atlas fail at 5.6% versus 20.7% inside. Severe failures are compact: $K = 5$ at 2 kcal mol^{-1} (5.5% of molecules) and $K = 2$ at 3 kcal mol^{-1} (2.9%), where the atlas holds 99.0% of failures. The failures that matter most are few and interpretable; the 1 kcal mol^{-1} tail is spread over a broad, ring-rich half of chemistry.

4.6 Conformational transfer

Errors are dominated by distortion. Over 3,112,861 conformations of 30,609 molecules, the minimised-geometry models predict relative conformational energies poorly and their error grows with distortion (Spearman 0.83; Appendix Fig. 6a): the median error is 2.8 kcal mol^{-1} within 5 kcal mol^{-1} of a molecule's lowest conformation and 48.2 kcal mol^{-1} beyond 40 kcal mol^{-1}. The hierarchy registers the departure: R/N_{atom} rises from 0.034 to 0.128 across distortion bands (Spearman 0.54; Appendix Fig. 6b), and frozen HCU ranks conformational errors with Spearman 0.474 ± 0.022 — by detecting distortion rather than chemistry.

The atlas largely does not transfer (Appendix Fig. 6c and Table 11). At matched distortion, 13 of 24 testable motifs keep a Holm-significant but small enrichment (error ratios ≤ 1.11, transfer ratios ≤ 0.22), and the strongest QM9 motifs show none: novel ring systems (3.32 on QM9, 1.00 on ANI-1x), sp^2 ring-fusion atoms (2.20, 0.99), twisted conjugation (2.46, 1.01). The motifs that keep an enrichment are strain and contact signatures that survive distortion (stretched bonds 1.11, short H$\cdots$H contacts 1.11, bridged polycycles 1.07) and some heteroatom motifs. The order-2 ratio term stays elevated on motif atoms for 17 of 24 motifs: the hierarchy's *response* transfers while its error consequences are swamped by distortion; rMD17 agrees (Appendix Table 10). The atlas and HCU thus describe minimised structures, as in screening; for dynamics with a minimised-geometry model the useful signal is the rise of R with distortion.

Cost. One size-extrapolating seed trained in 3.38 h (three runs sharing an H200). Inference with per-order outputs costs 0.049 ms per molecule (GPU); HCU adds 0.054 ms on one CPU core and needs no retraining (the latent distance needs stored training features and a GPU neighbour search, 0.072 ms); the grammar costs 0.85 ms per molecule. The ensemble multiplies training and inference by eight; HCU recovers 22% (with atlas indicators 58%) of its AUROC advantage over R for about the cost of a second forward pass.

5 Failure analysis and limitations

What the atlas does not capture. The atlas covers 75.8% of held-out molecules, and only 7.9% of failures lie outside it; those are mild (median 1.27 versus 1.58 kcal mol^{-1} inside; 99th percentile 3.8 versus 19.2 kcal mol^{-1}) and enriched in rings of seven or more atoms (lift 10.2), α,β-unsaturated carbonyls (3.5), esters and lactones (3.1) and imidates (2.8). With the hierarchy-blind candidates of Sec. 4.2 (large rings, C=N, N–N bonds) they form a second failure mode: every local environment is familiar and the decomposition converges, but the spatial arrangement of familiar groups (transannular contacts, conformational preferences) is new, and a sum of atom-centred terms can misassign it without any term becoming atypical. HCU flags 5.0% of these failures against 25.3% inside the atlas; the grammar makes this blind spot explicit.

Where the self-diagnosis fails. HCU almost never declares a large error safe: only 3 held-out molecules have a predicted error in the lowest quintile and an actual error above 2 kcal mol^{-1}. It fails in two other ways. It recognises but underestimates the extreme tail: the 15 largest errors (up to 59 kcal mol^{-1}) all get failure probabilities ≥ 0.74 but predicted mean errors ≤ 5.3 kcal mol^{-1}, an order of magnitude too small, because the model is fitted to a bulk in which errors rarely exceed 3 kcal mol^{-1}. And it raises false alarms on well-learned many-body chemistry: 2,393 molecules are in the top quintile of predicted error yet accurate to 0.5 kcal mol^{-1}; 84.9% carry an atlas motif and they are enriched in fluorine on unsaturated carbon (lift 9.0), fluorine α to N or O (8.7) and aromatic rings, environments where a large order-2 term is legitimate and was learned from the pool. Both expose HCU's central assumption — that a large or slowly decaying second order signals extrapolation

— which conflates "many-body but familiar" with "unfamiliar". Calibration is correspondingly not fully adaptive [24]: observed/predicted mean error is 0.97, 1.03 and 0.97 across size terciles but 1.02 inside and 0.78 outside the atlas, where HCU over-predicts by about a fifth.

Is the atlas an artefact of under-training? Every size-extrapolating run stopped by the paper's termination rule (1556 ± 100 epochs; the paper reports 1000–2000), and the code's accuracy gap to the paper where it reports values (at most 2%) is far smaller than the 29% cost of extrapolation or the motif error ratios. Models trained with the paper's initialisation are slightly more accurate out of distribution (0.713 ± 0.006 versus 0.736 ± 0.016 kcal mol^{-1}, same seeds) yet give the same atlas, mechanism and calibration; random-split models trained on 50,000 molecules keep the electronic motifs' error ratios (zwitterions 10.21); and topology and novelty motifs shrink with coverage while electronic motifs do not, a structure uniform under-training would not produce. The error ratios would shrink with more data; the identity of the motifs and the localisation of the breakdown in the second order would not.

Limitations. With two interaction blocks, order 2 is also the last order; more blocks could move or absorb the breakdown. The grammar is hand-written and QM9, a GDB-17 subset [25], is rich in strained cages and poor in drug-like functionality. Calibration is validated on formula-disjoint molecules of the same enumeration, not under arbitrary deployment shift, and no distribution-free guarantee is claimed. The ANI-1x comparison mixes functionals.

6 Related work

Uncertainty for machine-learned potentials. Ensemble disagreement dominates [14–16, 26–31] but is often over-confident out of distribution [32–34]. Alternatives include dropout [35, 36], stochastic-gradient MCMC [37], mean–variance and evidential heads [38–41], latent-space distances and densities [20, 42–44], last-layer approximations [45–47], secondary error predictors [48] and conformal prediction [49–52], whose exchangeability guarantee fails under deliberate covariate shift without corrections [53, 54]. No single-model method consistently matches ensembles [55–59]. HCU's inputs are the convergence diagnostics HIP-NN exposes by construction; its atypicality term shares the nearest-neighbour logic of latent-distance methods [20, 42] (in two hierarchy coordinates instead of the 80-dimensional feature space, which ranks errors better), and its post-hoc error model with isotonic recalibration [13] resembles error predictors [48]. What differs is that each input is tied to a failure of the many-body expansion, that no retraining is needed, and that calibration is evaluated on stoichiometry-disjoint chemistry together with a confirmed map of the structures that trigger it.

Failure analysis of molecular models. Out-of-distribution degradation is well documented [60–63], and small-to-large transfer underlies ANI-1 [64] and fragment-based learning [65]. The closest precedent [41] trained PhysNet with evidential regression on QM9 and identified poorly predicted groups (aliphatic nitro groups); we instead use the unmodified model's own decomposition, a pre-specified grammar confirmed on held-out stoichiometries under family-wise error control (subgroup discovery [66, 67] with sample splitting), a trace of each motif to the failing order and an off-equilibrium test. Per-atom energies are not unique [68, 69], hence our element-matched, seed-averaged contrasts. Hierarchical expansions are classical [70–72]; HIP-NN [5, 73] learns one, which makes its convergence observable.

7 Conclusions

Trained with its hierarchical penalty, HIP-NN's energy hierarchy records where the model extrapolates. Under size extrapolation HIP-NN fails in a structured way: 25 pre-specified motifs carry confirmed excess error in every seed, their atoms show a second-order term that grows and stops decaying, and the strained frameworks they describe are over-bound. A four-parameter, single-pass estimate built from the hierarchy's convergence is calibrated on held-out chemistry and beats R, although a distance in the model's feature space ranks errors better; severe failures are compact, the 1 kcal mol^{-1} tail is not, and on distorted geometries the hierarchy tracks distortion rather than chemistry. In practice: keep the hierarchical penalty, use the hierarchy to locate a failure and a feature-space distance to rank it, and treat cages with sp^2 ring-fusion atoms, unseen ring systems and charge-separated species as red flags; tensor sensitivities [73] and atlas-targeted active learning [16] are natural next tests.

References

[1] Raghunathan Ramakrishnan, Pavlo O. Dral, Matthias Rupp, and O. Anatole von Lilienfeld. Quantum chemistry structures and properties of 134 kilo molecules. *Scientific Data*, 1(1): 140022, 2014. doi: 10.1038/sdata.2014.22.

[2] K. T. Schütt, H. E. Sauceda, P.-J. Kindermans, A. Tkatchenko, and K.-R. Müller. SchNet – A deep learning architecture for molecules and materials. *The Journal of Chemical Physics*, 148 (24):241722, 2018. doi: 10.1063/1.5019779.

[3] Justin Gilmer, Samuel S. Schoenholz, Patrick F. Riley, Oriol Vinyals, and George E. Dahl. Neural message passing for quantum chemistry. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1263–1272. PMLR, 2017. URL https://proceedings.mlr.press/v70/gilmer17a.html.

[4] Oliver T. Unke and Markus Meuwly. PhysNet: A neural network for predicting energies, forces, dipole moments, and partial charges. *Journal of Chemical Theory and Computation*, 15(6): 3678–3693, 2019. doi: 10.1021/acs.jctc.9b00181.

[5] Nicholas Lubbers, Justin S. Smith, and Kipton Barros. Hierarchical modeling of molecular energies using a deep neural network. *The Journal of Chemical Physics*, 148(24):241715, 2018. doi: 10.1063/1.5011181.

[6] Nicholas Lubbers et al. hippynn. https://github.com/lanl/hippynn, 2019. Python library for atomistic machine learning. Los Alamos National Laboratory, LA-CC-19-093.

[7] David Rogers and Mathew Hahn. Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling*, 50(5):742–754, 2010. doi: 10.1021/ci100050t.

[8] Guy W. Bemis and Mark A. Murcko. The properties of known drugs. 1. Molecular frameworks. *Journal of Medicinal Chemistry*, 39(15):2887–2893, 1996. doi: 10.1021/jm9602928.

[9] Greg Landrum et al. RDKit: Open-source cheminformatics. https://www.rdkit.org, 2026. Release 2026_03_5 (Q1 2026), Zenodo. Concept DOI for all releases: 10.5281/zenodo.591637.

[10] H. B. Mann and D. R. Whitney. On a test of whether one of two random variables is stochastically larger than the other. *The Annals of Mathematical Statistics*, 18(1):50–60, 1947. doi: 10.1214/aoms/1177730491.

[11] Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979. URL https://www.jstor.org/stable/4615733.

[12] Bianca Zadrozny and Charles Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 694–699. ACM, 2002. doi: 10.1145/775047.775151.

[13] Volodymyr Kuleshov, Nathan Fenner, and Stefano Ermon. Accurate uncertainties for deep learning using calibrated regression. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 2796–2804. PMLR, 2018. URL https://proceedings.mlr.press/v80/kuleshov18a.html.

[14] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017. URL https://proceedings.neurips.cc/paper/2017/hash/9ef2ed4b7fd2c810847ffa5fa85bce38-Abstract.html.

[15] Andrew A. Peterson, Rune Christensen, and Alireza Khorshidi. Addressing uncertainty in atomistic machine learning. *Physical Chemistry Chemical Physics*, 19(18):10978–10985, 2017. doi: 10.1039/C7CP00375G.

[16] Justin S. Smith, Ben Nebgen, Nicholas Lubbers, Olexandr Isayev, and Adrian E. Roitberg. Less is more: Sampling chemical space with active learning. *The Journal of Chemical Physics*, 148 (24):241733, 2018. doi: 10.1063/1.5023802.

[17] Dan Levi, Liran Gispan, Niv Giladi, and Ethan Fetaya. Evaluating and calibrating uncertainty prediction in regression tasks. *Sensors*, 22(15):5540, 2022. doi: 10.3390/s22155540.

[18] Mahdi Pakdaman Naeini, Gregory Cooper, and Milos Hauskrecht. Obtaining well calibrated probabilities using Bayesian binning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 29, pages 2901–2907, 2015. doi: 10.1609/aaai.v29i1.9602.

[19] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330. PMLR, 2017. URL https://proceedings.mlr.press/v70/guo17a.html.

[20] Jon Paul Janet, Chenru Duan, Tzuhsiung Yang, Aditya Nandy, and Heather J. Kulik. A quantitative uncertainty metric controls error in neural network-driven chemical discovery. *Chemical Science*, 10(34):7913–7922, 2019. doi: 10.1039/C9SC02298H.

[21] Justin S. Smith, Roman Zubatyuk, Benjamin Nebgen, Nicholas Lubbers, Kipton Barros, Adrian E. Roitberg, Olexandr Isayev, and Sergei Tretiak. The ANI-1ccx and ANI-1x data sets, coupled-cluster and density functional theory properties for molecules. *Scientific Data*, 7 (1):134, 2020. doi: 10.1038/s41597-020-0473-z.

[22] Anders S. Christensen and O. Anatole von Lilienfeld. On the role of gradients for machine learning of molecular energies and forces. *Machine Learning: Science and Technology*, 1(4): 045018, 2020. doi: 10.1088/2632-2153/abba6f.

[23] Kenneth B. Wiberg. The concept of strain in organic chemistry. *Angewandte Chemie International Edition in English*, 25(4):312–322, 1986. doi: 10.1002/anie.198603121.

[24] Pascal Pernot. Calibration in machine learning uncertainty quantification: Beyond consistency to target adaptivity. *APL Machine Learning*, 1(4):046121, 2023. doi: 10.1063/5.0174943.

[25] Lars Ruddigkeit, Ruud van Deursen, Lorenz C. Blum, and Jean-Louis Reymond. Enumeration of 166 billion organic small molecules in the chemical universe database GDB-17. *Journal of Chemical Information and Modeling*, 52(11):2864–2875, 2012. doi: 10.1021/ci300415d.

[26] Christoph Schran, Krystof Brezina, and Ondrej Marsalek. Committee neural network potentials control generalization errors and enable active learning. *The Journal of Chemical Physics*, 153 (10):104105, 2020. doi: 10.1063/5.0016004.

[27] Félix Musil, Michael J. Willatt, Mikhail A. Langovoy, and Michele Ceriotti. Fast and accurate uncertainty estimation in chemical machine learning. *Journal of Chemical Theory and Computation*, 15(2):906–915, 2019. doi: 10.1021/acs.jctc.8b00959.

[28] Jesús Carrete, Hadrián Montes-Campos, Ralf Wanzenböck, Esther Heid, and Georg K. H. Madsen. Deep ensembles vs committees for uncertainty estimation in neural-network force fields: Comparison and application to active learning. *The Journal of Chemical Physics*, 158 (20):204801, 2023. doi: 10.1063/5.0146905.

[29] Justin S. Smith, Benjamin Nebgen, Nithin Mathew, Jie Chen, Nicholas Lubbers, Leonid Burakovsky, Sergei Tretiak, Hai Ah Nam, Timothy Germann, Saryu Fensin, and Kipton Barros. Automated discovery of a robust interatomic potential for aluminum. *Nature Communications*, 12(1):1257, 2021. doi: 10.1038/s41467-021-21376-0.

[30] Maksim Kulichenko, Kipton Barros, Nicholas Lubbers, Ying Wai Li, Richard Messerly, Sergei Tretiak, Justin S. Smith, and Benjamin Nebgen. Uncertainty-driven dynamics for active learning of interatomic potentials. *Nature Computational Science*, 3(3):230–239, 2023. doi: 10.1038/s43588-023-00406-5.

[31] Giulio Imbalzano, Yongbin Zhuang, Venkat Kapil, Kevin Rossi, Edgar A. Engel, Federico Grasselli, and Michele Ceriotti. Uncertainty estimation for molecular dynamics and sampling. *The Journal of Chemical Physics*, 154(7):074102, 2021. doi: 10.1063/5.0036522.

[32] Leonid Kahle and Federico Zipoli. Quality of uncertainty estimates from neural network potential ensembles. *Physical Review E*, 105(1):015311, 2022. doi: 10.1103/PhysRevE.105.015311.

[33] Glenn Palmer, Siqi Du, Alexander Politowicz, Joshua Paul Emory, Xiyu Yang, Anupraas Gautam, Grishma Gupta, Zhelong Li, Ryan Jacobs, and Dane Morgan. Calibration after bootstrap for accurate uncertainty quantification in regression models. *npj Computational Materials*, 8(1):115, 2022. doi: 10.1038/s41524-022-00794-8.

[34] Jonas Busk, Peter Bjørn Jørgensen, Arghya Bhowmik, Mikkel N. Schmidt, Ole Winther, and Tejs Vegge. Calibrated uncertainty for molecular property prediction using ensembles of message passing neural networks. *Machine Learning: Science and Technology*, 3(1):015012, 2022. doi: 10.1088/2632-2153/ac3eb3.

[35] Yarin Gal and Zoubin Ghahramani. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of the 33rd International Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 1050–1059, New York, New York, USA, 2016. PMLR. URL https://proceedings.mlr.press/v48/gal16.html.

[36] Mingjian Wen and Ellad B. Tadmor. Uncertainty quantification in molecular simulations with dropout neural network potentials. *npj Computational Materials*, 6(1):124, 2020. doi: 10.1038/s41524-020-00390-8.

[37] Stephan Thaler, Gregor Doehner, and Julija Zavadlav. Scalable Bayesian uncertainty quantification for neural network potentials: Promise and pitfalls. *Journal of Chemical Theory and Computation*, 19(14):4520–4532, 2023. doi: 10.1021/acs.jctc.2c01267.

[38] D. A. Nix and A. S. Weigend. Estimating the mean and variance of the target probability distribution. In *Proceedings of 1994 IEEE International Conference on Neural Networks (ICNN'94)*, volume 1, pages 55–60. IEEE, 1994. doi: 10.1109/ICNN.1994.374138.

[39] Alexander Amini, Wilko Schwarting, Ava Soleimany, and Daniela Rus. Deep evidential regression. In *Advances in Neural Information Processing Systems*, volume 33, pages 14927–14937. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper/2020/hash/aab085461de182608ee9f607f3f7d18f-Abstract.html.

[40] Ava P. Soleimany, Alexander Amini, Samuel Goldman, Daniela Rus, Sangeeta N. Bhatia, and Connor W. Coley. Evidential deep learning for guided molecular property prediction and discovery. *ACS Central Science*, 7(8):1356–1367, 2021. doi: 10.1021/acscentsci.1c00546.

[41] Luis Itza Vazquez-Salazar, Eric D. Boittier, and Markus Meuwly. Uncertainty quantification for predictions of atomistic neural networks. *Chemical Science*, 13(44):13068–13084, 2022. doi: 10.1039/D2SC04056E.

[42] Albert Zhu, Simon Batzner, Albert Musaelian, and Boris Kozinsky. Fast uncertainty estimates in deep learning interatomic potentials. *The Journal of Chemical Physics*, 158(16):164111, 2023. doi: 10.1063/5.0136574.

[43] Tom Wollschläger, Nicholas Gao, Bertrand Charpentier, Mohamed Amine Ketata, and Stephan Günnemann. Uncertainty estimation for molecules: Desiderata and methods. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 37133–37156. PMLR, 2023. URL https://proceedings.mlr.press/v202/wollschlager23a.html.

[44] Joseph Musielewicz, Janice Lan, Matt Uyttendaele, and John R. Kitchin. Improved uncertainty estimation of graph neural network potentials using engineered latent space distances. *The Journal of Physical Chemistry C*, 128(49):20799–20810, 2024. doi: 10.1021/acs.jpcc.4c04972.

[45] Matthias Kellner and Michele Ceriotti. Uncertainty quantification by direct propagation of shallow ensembles. *Machine Learning: Science and Technology*, 5(3):035006, 2024. doi: 10.1088/2632-2153/ad594a.

[46] Filippo Bigi, Sanggyu Chong, Michele Ceriotti, and Federico Grasselli. A prediction rigidity formalism for low-cost uncertainties in trained neural networks. *Machine Learning: Science and Technology*, 5(4):045018, 2024. doi: 10.1088/2632-2153/ad805f.

[47] Jenna A. Bilbrey, Jesun S. Firoz, Mal-Soon Lee, and Sutanay Choudhury. Uncertainty quantification for neural network potential foundation models. *npj Computational Materials*, 11(1): 109, 2025. doi: 10.1038/s41524-025-01572-y.

[48] Salem Lahlou, Moksh Jain, Hadi Nekoei, Victor I. Butoi, Paul Bertin, Jarrid Rector-Brooks, Maksym Korablyov, and Yoshua Bengio. DEUP: Direct epistemic uncertainty prediction. *Transactions on Machine Learning Research*, 2023. URL https://openreview.net/forum?id=eGLdVRvvfQ.

[49] Vladimir Vovk, Alex Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005. ISBN 0387001522. doi: 10.1007/b106715.

[50] Anastasios N. Angelopoulos and Stephen Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023. doi: 10.1561/2200000101.

[51] Fredrik Svensson, Natalia Aniceto, Ulf Norinder, Isidro Cortes-Ciriano, Ola Spjuth, Lars Carlsson, and Andreas Bender. Conformal regression for quantitative structure–activity relationship modeling—quantifying prediction uncertainty. *Journal of Chemical Information and Modeling*, 58(5):1132–1140, 2018. doi: 10.1021/acs.jcim.8b00054.

[52] Yuge Hu, Joseph Musielewicz, Zachary W. Ulissi, and Andrew J. Medford. Robust and scalable uncertainty estimation with conformal prediction for machine-learned interatomic potentials. *Machine Learning: Science and Technology*, 3(4):045028, 2022. doi: 10.1088/2632-2153/aca7b1.

[53] Ryan J. Tibshirani, Rina Foygel Barber, Emmanuel Candès, and Aaditya Ramdas. Conformal prediction under covariate shift. In *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc., 2019. URL https://proceedings.neurips.cc/paper/2019/hash/8fb21ee7a2207526da55a679f0332de2-Abstract.html.

[54] Maxime Cauchois, Suyash Gupta, Alnur Ali, and John C. Duchi. Robust validation: Confident predictions even when distributions shift. *Journal of the American Statistical Association*, 119 (548):3033–3044, 2024. doi: 10.1080/01621459.2023.2298037.

[55] Aik Rui Tan, Shingo Urata, Samuel Goldman, Johannes C. B. Dietschreit, and Rafael Gómez-Bombarelli. Single-model uncertainty quantification in neural network potentials does not consistently outperform model ensembles. *npj Computational Materials*, 9(1):225, 2023. doi: 10.1038/s41524-023-01180-8.

[56] Lior Hirschfeld, Kyle Swanson, Kevin Yang, Regina Barzilay, and Connor W. Coley. Uncertainty quantification using neural networks for molecular property prediction. *Journal of Chemical Information and Modeling*, 60(8):3770–3780, 2020. doi: 10.1021/acs.jcim.0c00502.

[57] Gabriele Scalia, Colin A. Grambow, Barbara Pernici, Yi-Pei Li, and William H. Green. Evaluating scalable uncertainty estimation methods for deep learning-based molecular property prediction. *Journal of Chemical Information and Modeling*, 60(6):2697–2717, 2020. doi: 10.1021/acs.jcim.9b00975.

[58] Esther Heid, Charles J. McGill, Florence H. Vermeire, and William H. Green. Characterizing uncertainty in machine learning for chemistry. *Journal of Chemical Information and Modeling*, 63(13):4012–4029, 2023. doi: 10.1021/acs.jcim.3c00373.

[59] Maria H. Rasmussen, Chenru Duan, Heather J. Kulik, and Jan H. Jensen. Uncertain of uncertainties? A comparison of uncertainty quantification metrics for chemical data sets. *Journal of Cheminformatics*, 15(1):121, 2023. doi: 10.1186/s13321-023-00790-0.

[60] Marta Glavatskikh, Jules Leguy, Gilles Hunault, Thomas Cauchy, and Benoit Da Mota. Dataset's chemical diversity limits the generalizability of machine learning predictions. *Journal of Cheminformatics*, 11(1):69, 2019. doi: 10.1186/s13321-019-0391-2.

[61] Prudencio Tossou, Cas Wognum, Michael Craig, Hadrien Mary, and Emmanuel Noutahi. Real-world molecular out-of-distribution: Specification and investigation. *Journal of Chemical Information and Modeling*, 64(3):697–711, 2024. doi: 10.1021/acs.jcim.3c01774.

[62] Joshua A. Vita and Daniel Schwalbe-Koda. Data efficiency and extrapolation trends in neural network interatomic potentials. *Machine Learning: Science and Technology*, 4(3):035031, 2023. doi: 10.1088/2632-2153/acf115.

[63] Sina Stocker, Johannes Gasteiger, Florian Becker, Stephan Günnemann, and Johannes T. Margraf. How robust are modern graph neural network potentials in long and hot molecular dynamics simulations? *Machine Learning: Science and Technology*, 3(4):045010, 2022. doi: 10.1088/2632-2153/ac9955.

[64] J. S. Smith, O. Isayev, and A. E. Roitberg. ANI-1: an extensible neural network potential with DFT accuracy at force field computational cost. *Chemical Science*, 8(4):3192–3203, 2017. doi: 10.1039/C6SC05720A.

[65] Bing Huang and O. Anatole von Lilienfeld. Quantum machine learning using atom-in-molecule-based fragments selected on the fly. *Nature Chemistry*, 12(10):945–951, 2020. doi: 10.1038/s41557-020-0527-z.

[66] Willi Klösgen. Explora: A multipattern and multistrategy discovery assistant. In Usama M. Fayyad, Gregory Piatetsky-Shapiro, Padhraic Smyth, and Ramasamy Uthurusamy, editors, *Advances in Knowledge Discovery and Data Mining*, pages 249–271. AAAI/MIT Press, 1996.

[67] Francisco Herrera, Cristóbal José Carmona, Pedro González, and María José del Jesus. An overview on subgroup discovery: foundations and applications. *Knowledge and Information Systems*, 29(3):495–525, 2011. doi: 10.1007/s10115-010-0356-2.

[68] Sanggyu Chong, Federico Grasselli, Chiheb Ben Mahmoud, Joe D. Morrow, Volker L. Deringer, and Michele Ceriotti. Robustness of local predictions in atomistic machine learning models. *Journal of Chemical Theory and Computation*, 19(22):8020–8031, 2023. doi: 10.1021/acs.jctc.3c00704.

[69] Xin Chen, Mathias S. Jørgensen, Jun Li, and Bjørk Hammer. Atomic energies from a convolutional neural network. *Journal of Chemical Theory and Computation*, 14(7):3933–3942, 2018. doi: 10.1021/acs.jctc.8b00149.

[70] Frank H. Stillinger and Thomas A. Weber. Computer simulation of local order in condensed phases of silicon. *Physical Review B*, 31(8):5262–5271, 1985. doi: 10.1103/PhysRevB.31.5262.

[71] Kun Yao, John E. Herr, and John Parkhill. The many-body expansion combined with neural networks. *The Journal of Chemical Physics*, 146(1):014106, 2017. doi: 10.1063/1.4973380.

[72] Ryan M. Richard, Ka Un Lao, and John M. Herbert. Understanding the many-body expansion for large systems. I. Precision considerations. *The Journal of Chemical Physics*, 141(1):014108, 2014. doi: 10.1063/1.4885846.

[73] Michael Chigaev, Justin S. Smith, Steven Anaya, Benjamin Nebgen, Matthew Bettencourt, Kipton Barros, and Nicholas Lubbers. Lightweight and effective tensor sensitivity for atomistic neural networks. *The Journal of Chemical Physics*, 158(18):184108, 2023. doi: 10.1063/5.0142127.

A Reproducibility and logs

All numbers in this paper are generated from log files by `code/make_macros.py` and `code/make_tables.py`; each macro in `logs/results_macros.tex` carries the path of the log it was computed from, and every paper-as-reported value is listed with its table or section of origin in `logs/paper_reference.md`. Per-seed training records (hyper-parameters, seeds, split indices, per-epoch training/validation MAE with time stamps, checkpoints every 250 epochs, the best-validation model, per-molecule predictions with per-atom per-order energies and R) are in `logs/runs/<run>/`. Motif assignments for every nine-heavy-atom molecule

are in `logs/motifs/motif_assignments_test9.csv.gz`; the discovery/evaluation partition in `logs/partition/`; the hashed pre-registration and fitted uncertainty models in `logs/discovery/`; held-out results in `logs/confirm/`. Table 3 is generated from the `config.json` of all 30 runs, which the generator checks to be identical apart from $\lambda_R = 0$ in the three ablation runs (the initialisation scheme of the diagnostic runs is recorded in their `config.json`).

Table 3: HIP-NN hyper-parameters as logged by every run, with the location of the corresponding value in Ref. [5].

Hyper-parameter	Value (all runs)	Paper source
Interaction blocks N_{int}	2	Sec. II.6
On-site layers per block $N_{\text{on-site}}$	3	Sec. II.6
Features per layer N_{feature}	80	Sec. II.6, Table II
Sensitivity functions	20	Sec. II.6
R_{low} / R_{high} / R_{cut} (Bohr)	1.7 / 10.0 / 15.0	Sec. II.6
λ_{L2} / λ_R	10^{-6} / 10^{-2}	Sec. II.6
Minibatch size	30	Sec. II.6
Adam η_{init}, β_1, β_2	10^{-3}, 0.9, 0.999	Sec. II.6
α_{decay} / t_{patience} / t_{init} / $t_{\max}$	0.5 / 50 / 100 / 2000	Sec. II.5.2, II.6
Validation molecules	1000	Sec. II.6
Order-n read-out rescaling at init.	$(0.01)^n$	Sec. II.6
Parameters (trainable)	234,327 (234,322)	Table II: 234k

B Per-seed training results

Table 4: Per-run training summary: epochs until the paper's termination rule fired (or $t_{\max}$), number of learning-rate halvings, best-validation epoch, wall-clock training time (up to three concurrent runs per H200 GPU), validation and test MAE (kcal mol^{-1}) and fraction of test errors above 1 kcal mol^{-1}.

Protocol	seed	epochs	halvings	best ep.	hours	val. MAE	test MAE	test RMSE	test >1 (%)
$\leq$8 heavy	0	1622	12	1521	3.5	0.286	0.785	2.080	17.8
$\leq$8 heavy	1	1277	12	1176	2.8	0.299	0.734	1.731	17.9
$\leq$8 heavy	2	1766	11	1665	3.9	0.307	0.787	2.164	17.7
$\leq$8 heavy	3	1720	11	1619	3.8	0.300	0.777	2.055	17.9
$\leq$8 heavy	4	1702	10	1601	3.6	0.294	0.766	2.107	16.6
$\leq$8 heavy	5	1133	8	1032	2.5	0.282	0.746	1.907	17.3
$\leq$8 heavy	6	1926	12	1825	4.1	0.300	0.738	1.965	17.0
$\leq$8 heavy	7	1302	10	1201	2.9	0.430	0.778	2.039	18.4
$\leq$8 heavy, λ_R=0	0	1582	10	1481	3.3	0.310	0.764	1.796	18.6
$\leq$8 heavy, λ_R=0	1	1872	11	1771	3.9	0.299	0.767	1.910	18.3
$\leq$8 heavy, λ_R=0	2	1551	12	1450	3.3	0.360	0.822	2.001	20.1
$\leq$8 heavy, paper init.	0	1339	10	1238	2.9	0.271	0.755	1.911	17.3
$\leq$8 heavy, paper init.	1	1503	11	1402	3.1	0.297	0.739	1.826	17.3
$\leq$8 heavy, paper init.	2	1395	12	1294	2.9	0.301	0.738	1.909	16.9
random 110,426	0	1284	11	1183	12.2	0.238	0.263	0.570	2.2
random 110,426	1	1643	12	1542	15.5	0.245	0.246	0.508	2.1
random 110,426	2	1752	12	1651	16.4	0.279	0.257	0.529	2.3
random 110,426	3	1242	11	1141	12.0	0.271	0.282	0.528	2.8
random 50,000	0	1509	11	1408	7.4	0.307	0.333	0.762	4.6
random 50,000	1	1283	10	1182	6.4	0.346	0.341	0.713	4.5
random 50,000	2	1308	10	1207	6.5	0.384	0.360	0.767	5.2
random 50,000, paper init.	0	1562	10	1461	7.0	0.328	0.347	0.733	4.8
random 50,000, paper init.	1	1869	13	1768	8.1	0.370	0.357	0.757	5.3
random 50,000, paper init.	2	1629	11	1528	7.2	0.367	0.353	0.837	5.0
random 21,799	0	1053	11	952	2.3	0.488	0.561	1.123	13.1
random 21,799	1	1397	10	1296	2.9	0.549	0.539	1.141	11.8
random 21,799	2	1290	9	1189	2.8	0.572	0.536	1.165	11.6
random 21,799, paper init.	0	1193	10	1092	2.7	0.481	0.542	1.119	12.2
random 21,799, paper init.	1	1269	9	1168	2.8	0.550	0.527	1.110	11.2
random 21,799, paper init.	2	1208	9	1107	2.7	0.618	0.579	1.177	13.7

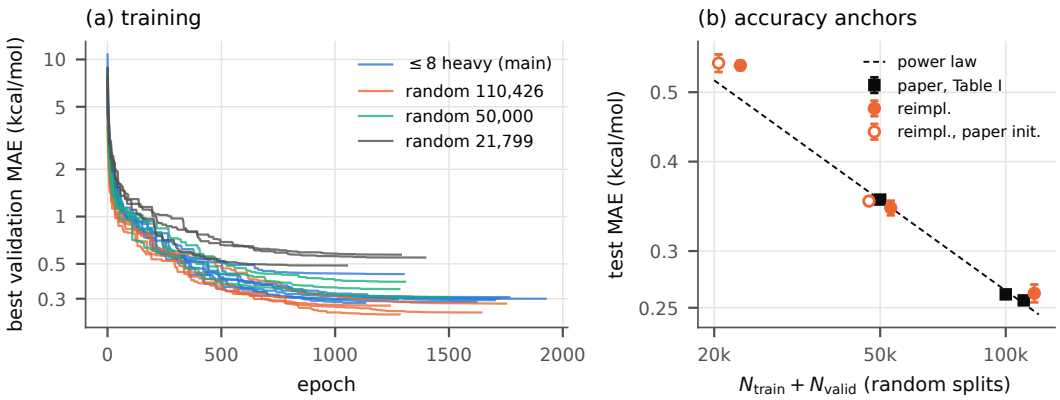


Figure 3: (a) Best validation MAE during training for the size-extrapolating models and the random-split anchors (random-split validation sets contain nine-heavy-atom molecules). (b) Test MAE of the random-split anchors (mean ± s.e. over seeds; open markers: the paper's initialisation) against Table I of Ref. [5] and a power law through its three points.

Fidelity and initialisation. The paper initialises all weights Glorot-uniform with zero biases; hippynn uses Xavier-normal weights (with PyTorch's fan convention for the three-index interaction tensor) and PyTorch's defaults for the ResNet output layers and the biases. To test whether this difference matters we retrained, with the paper's scheme and identical splits, three seeds at 21,799 and 50,000 random-split molecules and three size-extrapolating seeds (runs `*_pinit_*`). The paired differences (paper minus default initialisation) of the test MAE are 0.004 ± 0.020 and 0.008 ± 0.007 kcal mol^{-1} on the random splits and -0.025 ± 0.016 kcal mol^{-1} on all nine-heavy-atom molecules for the size-extrapolating models. Re-applied without any re-selection, the frozen atlas and the HCU protocol give the same conclusions for these models (Table 9): all 25 motifs keep a significant excess error and 25 an order-2 excess at their atoms, the error ratios correlate with those of the main models at Spearman 0.99, $K = 10$, 5 and 1 at 1, 2 and 3 kcal mol^{-1}, and HCU reaches $\mathrm{ECE}_{\mathrm{rel}} = 0.034 \pm 0.006$ (largest bin deviation 7.0%) and AUROC 0.771 ± 0.006.

C Candidate motifs

Table 5: All 68 pre-specified candidate motifs. Coverage = percentage of molecules matching, in the ≤8-heavy-atom pool and among the 9-heavy-atom molecules. SMARTS are matched on the molecular graph with explicit hydrogens.

ID	Name	Definition (SMARTS or rule)	≤8 %	9-heavy %
S01	3-membered ring	`[r3]`	37.0	42.7
S02	epoxide/aziridine	`[#7,#8;r3]`	19.0	21.8
S03	unsaturated 3-ring	`[r3]=*`	0.0	0.0
S04	4-membered ring	`[r4]`	31.0	38.4
S05	oxetane/azetidine	`[#7,#8;r4]`	16.4	20.1
S06	unsaturated 4-ring	`[r4]=*`	7.7	9.1
S07	ring of ≥7 atoms	`[r7,r8,r9]`	1.2	3.2
E01	aromatic ring	`a`	18.0	16.8
E02	5-ring heteroaromatic	`[n,o;r5;a]`	13.3	11.9
E03	6-ring aromatic	`[a;r6]`	4.5	4.6
E04	conjugated diene	`[#6;!a]=[#6;!a]-[#6;!a]=[#6;!a]`	0.2	0.4
E05	enone/enal	`[#6;!a]=[#6;!a]-[#6]=[#8]`	1.5	2.2
E06	cumulene	`[#6,#7,#8]=[#6X2]=[#6,#7,#8]`	0.0	0.0
E07	C≡C triple bond	`[#6]#[#6]`	13.1	13.5
E08	nitrile	`[#6]#[#7]`	12.0	12.7
E09	conjugated triple bond	`*#*-[$(*=*),$(*#*),$(a)]`	4.9	4.6
E10	enamine	`[NX3;!a]-[#6;!a]=[#6;!a]`	0.8	1.3

ID	Name	Definition	≤8 %	9-heavy %
E11	enol ether	`[OX2;!a]-[#6;!a]=[#6;!a]`	0.0	0.0
E12	push-pull enone	`[NX3,OX2;!a]-[#6;!a]=[#6;!a]-[#6]=[O,N]`	0.4	0.6
E13	non-aromatic C=N	`[#6;!a]=[#7;!a]`	10.2	12.5
E14	amidine/guanidine	`[#7]-[#6;!a]=[#7;!a]`	3.0	4.1
E15	imidate	`[#8]-[#6;!a]=[#7;!a]`	6.4	7.6
E16	amide/lactam	`[NX3][CX3](=[OX1])`	10.1	10.6
E17	ester/lactone	`[OX2][CX3](=[OX1])`	5.1	5.4
E18	carboxylic acid	`[OX2H1][CX3](=[OX1])`	0.0	0.0
E19	urea/carbamate	`[#7,#8][CX3](=[OX1])[#7,#8]`	1.8	1.8
E20	1,3-dicarbonyl/imide	`[CX3](=[O,N])[#6,#7][CX3]=[O,N]`	3.2	3.5
E21	aldehyde	`[CX3H1](=O)`	13.4	13.8
E22	ketone	`[#6][CX3](=O)[#6]`	9.8	11.7
E23	zwitterion	`[+,-]`	0.3	0.5
H01	N-N bond	`[#7]~[#7]`	6.6	5.4
H02	N-O bond	`[#7]~[#8]`	5.1	4.6
H03	O-O bond	`[#8]~[#8]`	0.0	0.0
H04	anomeric sp^3 C	`[CX4]([#7,#8,#9])[#7,#8,#9]`	2.3	2.9
H05	C bearing 3 N/O/F	`[#6]([#7,#8,#9])([#7,#8,#9])[#7,#8,#9]`	2.6	2.2
H06	hemiacetal	`[CX4]([OX2H1])[#7,#8]`	0.0	0.0
H07	fluorine	`[#9]`	1.4	1.5
H08	CF2/CF3	`[#6]([#9])[#9]`	0.4	0.3
H09	F on unsaturated C	`[#9][#6;X3,X2,a]`	1.0	1.2
H10	F alpha to N/O	`[#9][#6][#7,#8]`	0.7	0.7
H11	N/O-substituted nitrile	`[#7,#8]-[#6]#[#7]`	0.0	0.0
H12	tertiary amine	`[NX3;H0;!$(N-[#6]=[#7,#8]);!$(N-a);!$(N-[#6]#*);!$(N~[#7,#8])]`	8.2	11.2
H13	primary amine	`[NX3;H2]`	8.9	8.5
H14	hydroxyl	`[OX2H1]`	30.2	33.4
S08	fused small ring	atom shared by two SSSR rings, one of size ≤4	31.7	44.6
S09	bridged polycycle	two SSSR rings sharing ≥3 atoms (atoms = shared atoms)	8.3	13.1
S10	spiro centre	two SSSR rings sharing exactly one atom	4.5	8.7
S11	cage (≥3 rings)	fused/bridged/spiro ring system containing ≥3 SSSR rings	14.4	23.9
S12	sp^2 ring-fusion atom	atom in ≥2 SSSR rings carrying a double bond	2.7	5.1
S13	fused bicyclic	two SSSR rings sharing exactly two atoms (one bond)	25.6	38.7
S14	acyclic	no ring	16.7	8.9
S15	quaternary sp^3 C	sp^3 carbon with four heavy-atom neighbours	29.3	42.8
H15	≥3 N atoms	≥3 nitrogen atoms	9.2	10.0
H16	≥3 O atoms	≥3 oxygen atoms	6.9	11.1
G01	intramolecular H-bond	polar H (on N/O) to N/O/F acceptor ≥3 bonds from donor: H$\cdots$A<2.5 Å, D–H$\cdots$A>110°	7.4	9.8
G02	short H$\cdots$H contact	two H on different heavy atoms closer than 2.0 Å	0.8	1.2
G03	compressed 1,3 contact	heavy atoms two bonds apart closer than 2.05 Å	9.0	11.5
G04	stretched bond >1.58 Å	heavy-atom bond longer than 1.58 Å	4.4	8.2
G05	sp^2 angle <100°	sp^2 heavy atom with a bond angle <100°	9.6	11.6

ID	Name	Definition	$\leq$8 %	9-heavy %
G06	non-planar amide N	non-aromatic N with 3 neighbours, attached to C=X, sum of angles $<$350°	2.9	3.8
G07	twisted conjugation	single bond between two π-bonded centres (not in 3/4-rings), dihedral $>$30° from planarity	0.1	0.4
G08	eclipsed H-C-C-H	H–C–C–H dihedral $<$25° across an sp^3–sp^3 C–C bond	44.9	50.6
N01	novel radius-1 environment	radius-1 Morgan environment absent from all $\leq$8-heavy-atom molecules	0.0	11.8
N02	novel radius-2 environment	radius-2 Morgan environment absent from all $\leq$8-heavy-atom molecules	0.0	87.5
N03	novel ring system	ring system (canonical SMILES) absent from all $\leq$8-heavy-atom molecules	0.0	9.5
N04	novel Murcko scaffold	Bemis–Murcko scaffold absent from all $\leq$8-heavy-atom molecules	0.0	12.6
N05	rare radius-2 environment	radius-2 Morgan environment present in $<$5 $\leq$8-heavy-atom molecules	94.3	99.0
N06	novel radius-3 environment	radius-3 Morgan environment absent from all $\leq$8-heavy-atom molecules	0.0	96.6

D Additional analyses

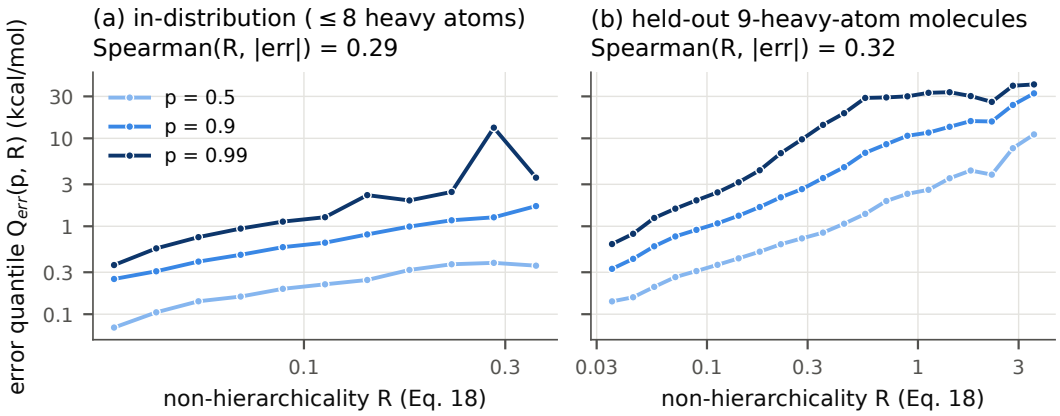


Figure 4: Error quantiles conditioned on the non-hierarchicality R (the analogue of Fig. 3 of Ref. [5]), pooled over the eight size-extrapolating models, for (a) their in-distribution validation molecules and (b) the held-out nine-heavy-atom evaluation molecules (bins of 0.1 decades with at least 30 molecules).

In-distribution control. Table 8 reports, for the held-out nine-heavy-atom molecules, the error ratio of every atlas motif under the size-extrapolating models and under random-split models that saw nine-heavy-atom molecules during training (each random-split model is evaluated only on the evaluation-partition molecules in its own test set).

Hierarchical-regulariser ablation and in-distribution informativeness of R. Table 9 compares, for seeds 0–2, the size-extrapolating models trained with and without the hierarchical penalty, and reports the same statistics of R for the random-split models on their own (in-distribution) test sets.

Revised MD17. With only ten molecules no motif-level inference is possible, so rMD17 serves as a descriptive check of the conformational-transfer conclusion (Table 10). All conformations are used and errors are relative to each molecule's lowest-energy conformation, as for ANI-1x. The minimised-geometry models get the relative energies of MD conformations badly wrong: the median ratio of error to distortion energy is 0.54–1.02 across molecules, with median errors of 4.1–21.4 kcal mol^{-1}

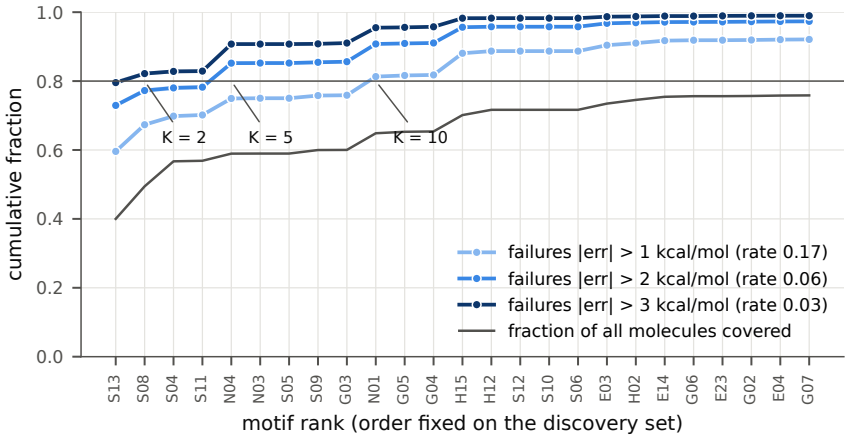


Figure 5: Cumulative failure attribution of the atlas motifs in the order fixed on discovery data, with motif identities (detailed version of Fig. 2c; means over the eight seeds, held-out molecules); grey: fraction of all held-out molecules in the union.

at median distortions of 7.7–21.5 kcal mol^{-1}, and neither R nor HCU ranks conformations within a molecule (Spearman correlations between -0.21 and 0.23). The five molecules within the training size range (ethanol, malonaldehyde, benzene, toluene, uracil) behave like the larger ones.

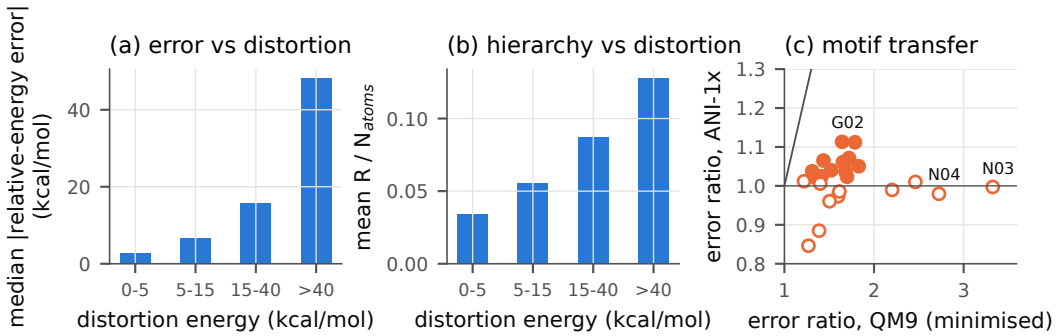


Figure 6: Transfer to ANI-1x conformations. (a) Median error of the predicted relative conformational energy and (b) mean R/N_{atom} against the distortion energy above each molecule's lowest sampled conformation; (c) distortion-adjusted error ratio on ANI-1x against the held-out QM9 ratio for each atlas motif (geometric means; filled: Holm-significant transfer; diagonal: full transfer).

ANI-1x transfer per motif. Table 11 lists the transfer statistics of every testable atlas motif. Error ratios on QM9 and ANI-1x are geometric-mean ratios (the transfer ratio TR is the ratio of their logarithms); "ANI" is the distortion-adjusted primary outcome, "raw" the unadjusted mean conformational error; the four band columns repeat the test within distortion-energy bands; $\Delta r^{(2)}$ is the element-matched excess of the order-2 ratio term on the motif atoms, averaged over all conformations.

Table 6: Statistics of every candidate. Discovery columns: number of discovery molecules carrying the candidate, seed-averaged error ratio and R/N_{atom} ratio used by the selection rule (atlas = selected). Evaluation columns: held-out error ratio and Holm-adjusted p over all testable candidates (sensitivity family).

		coverage %		discovery partition				evaluation partition	
ID	Candidate	$\leq$8 pool	9-heavy	n	ER	R ratio	atlas	ER	p_{Holm} (all)
S01	3-membered ring	37.0	44.3	17986	0.93	0.95		0.85	1.000
S02	epoxide/aziridine	19.0	23.0	8940	0.87	0.91		0.92	1.000
S03	unsaturated 3-ring	0.0	0.0	0	–	–		–	–
S04	4-membered ring	31.0	39.9	16123	1.53	1.46	✓	1.36	10^{-192}
S05	oxetane/azetidine	16.4	21.1	8293	1.60	1.46	✓	1.58	10^{-272}
S06	unsaturated 4-ring	7.7	8.8	4204	1.39	1.59	✓	1.50	10^{-71}
S07	ring of $\geq$7 atoms	1.2	3.3	1375	1.78	1.01		1.84	$<10^{-300}$
E01	aromatic ring	18.0	15.9	8016	1.22	1.34		1.40	$<10^{-300}$
E02	5-ring heteroaromatic	13.3	11.3	5710	1.17	1.10		1.37	10^{-183}
E03	6-ring aromatic	4.5	4.3	2228	1.36	1.78	✓	1.24	10^{-127}
E04	conjugated diene	0.2	0.2	245	1.98	1.63	✓	1.68	10^{-9}
E05	enone/enal	1.5	2.1	1060	1.12	1.28		1.00	10^{-7}
E06	cumulene	0.0	0.0	0	–	–		–	–
E07	C$\equiv$C triple bond	13.1	12.0	7005	0.54	0.87		0.54	1.000
E08	nitrile	12.0	13.1	5431	0.64	0.83		0.69	1.000
E09	conjugated triple bond	4.9	3.9	2506	0.73	1.19		0.72	1.000
E10	enamine	0.8	1.1	642	1.13	1.12		1.42	10^{-45}
E11	enol ether	0.0	0.0	4	–	–		–	–
E12	push-pull enone	0.4	0.6	344	1.21	1.19		1.54	10^{-27}
E13	non-aromatic C=N	10.2	12.0	5855	1.18	1.01		1.40	10^{-163}
E14	amidine/guanidine	3.0	4.0	1820	1.54	1.11	✓	1.91	10^{-222}
E15	imidate	6.4	7.2	3620	0.95	0.91		1.13	10^{-24}
E16	amide/lactam	10.1	9.7	5305	0.75	0.87		0.92	10^{-21}
E17	ester/lactone	5.1	5.4	2419	0.78	0.94		0.97	10^{-4}
E18	carboxylic acid	0.0	0.0	0	–	–		–	–
E19	urea/carbamate	1.8	1.5	998	0.76	0.88		1.04	10^{-18}
E20	1,3-dicarbonyl/imide	3.2	3.5	1523	0.97	0.94		0.97	10^{-39}
E21	aldehyde	13.4	14.0	5960	0.68	0.78		0.69	1.000
E22	ketone	9.8	11.9	5089	0.85	0.93		0.79	1.000
E23	zwitterion	0.3	0.4	256	10.74	4.82	✓	10.54	10^{-120}
H01	N-N bond	6.6	4.8	2751	1.15	1.05		1.49	10^{-221}
H02	N-O bond	5.1	4.3	2219	1.33	1.30	✓	1.44	10^{-130}
H03	O-O bond	0.0	0.0	0	–	–		–	–
H04	anomeric sp^3 C	2.3	3.0	1213	1.03	0.90		1.21	10^{-48}
H05	C bearing 3 N/O/F	2.6	1.8	1279	1.10	1.20		1.23	10^{-35}
H06	hemiacetal	0.0	0.0	0	–	–		–	–
H07	fluorine	1.4	1.1	921	0.91	1.25		0.88	0.265
H08	CF2/CF3	0.4	0.2	206	1.03	0.83		1.00	1.000
H09	F on unsaturated C	1.0	0.8	715	0.88	1.37		0.85	0.404
H10	F alpha to N/O	0.7	0.5	462	0.86	1.38		0.89	0.064
H11	N/O-substituted nitrile	0.0	0.0	0	–	–		–	–
H12	tertiary amine	8.2	11.6	4646	1.40	1.27	✓	1.44	10^{-180}
H13	primary amine	8.9	7.7	4362	0.97	0.99		1.15	10^{-56}
H14	hydroxyl	30.2	33.3	14872	0.59	0.73		0.64	1.000
S08	fused small ring	31.7	46.2	18766	1.78	1.61	✓	1.57	$<10^{-300}$
S09	bridged polycycle	8.3	13.5	5530	2.44	1.88	✓	2.23	$<10^{-300}$
S10	spiro centre	4.5	9.1	3605	2.70	2.05	✓	2.21	10^{-145}
S11	cage ($\geq$3 rings)	14.4	24.6	10126	2.65	2.23	✓	2.28	$<10^{-300}$
S12	sp^2 ring-fusion atom	2.7	4.7	2543	2.87	2.60	✓	2.77	$<10^{-300}$
S13	fused bicyclic	25.6	40.0	16332	2.24	1.83	✓	1.97	$<10^{-300}$
S14	acyclic	16.7	7.5	4893	0.55	0.62		0.54	1.000
S15	quaternary sp^3 C	29.3	44.4	18054	1.14	1.17		1.05	1.000
H15	$\geq$3 N atoms	9.2	9.1	4975	1.30	1.15	✓	1.60	$<10^{-300}$
H16	$\geq$3 O atoms	6.9	9.5	6003	0.77	0.86		0.85	10^{-4}
G01	intramolecular H-bond	7.4	9.2	4689	0.83	0.80		0.92	0.139
G02	short H$\cdots$H contact	0.8	1.4	458	1.75	1.73	✓	1.59	10^{-84}
G03	compressed 1,3 contact	9.0	11.8	4925	2.39	2.29	✓	2.19	$<10^{-300}$
G04	stretched bond $>$1.58 Å	4.4	8.0	3752	3.03	2.65	✓	2.70	$<10^{-300}$
G05	sp^2 angle $<$100°	9.6	11.5	5298	1.66	1.61	✓	1.78	10^{-199}
G06	non-planar amide N	2.9	3.6	1791	1.26	1.17	✓	1.94	10^{-195}
G07	twisted conjugation	0.1	0.3	208	3.14	1.96	✓	2.79	10^{-40}
G08	eclipsed H-C-C-H	44.9	52.9	21094	1.10	1.00		0.97	10^{-6}
N01	novel radius-1 environment	0.0	11.5	5482	2.01	1.52	✓	1.89	$<10^{-300}$
N02	novel radius-2 environment	0.0	86.8	39353	1.46	1.61		1.25	10^{-23}
N03	novel ring system	0.0	8.9	4547	4.99	2.38	✓	4.44	$<10^{-300}$
N04	novel Murcko scaffold	0.0	12.3	5775	4.29	2.13	✓	3.71	$<10^{-300}$
N05	rare radius-2 environment	94.3	99.0	44015	1.18	1.48		0.98	1.000
N06	novel radius-3 environment	0.0	96.6	42946	1.07	1.02		0.99	0.139

Table 7: Uncertainty estimates on held-out molecules (means over eight seeds, ± s.e. where shown; all fitted and calibrated on discovery molecules). Models: trained HIP-NN models required. Max dev.: largest relative deviation of a calibration bin; prob. ECE: calibration error of $P(|\varepsilon| > 1\ \text{kcal mol}^{-1})$; ρ: Spearman correlation with $|\varepsilon|$; Prec./Rec.: flagging with $P \geq 0.5$; cov_{90}: coverage of nominal 90% bounds. Rows below the second rule were added after the confirmatory analysis; $\bar{L}$: mean nearest-neighbour atypicality of the atoms in the 80-dimensional final-layer feature space (Sec. 3.3).

Predictor	models	ECE_{rel}	max dev. %	prob. ECE	ρ	AUROC	Prec.	Rec.	cov_{90} %
R only (paper signal)	1	0.034 ± 0.004	7.9	0.020	0.331	0.745 ± 0.006	0.59	0.21	90.0
HCU (this work)	1	0.042 ± 0.004	9.3	0.017	0.367	0.770 ± 0.005	0.61	0.24	89.6
HCU + atlas indicators	1	0.028 ± 0.003	6.4	0.009	0.419	0.810 ± 0.003	0.66	0.30	89.6
ensemble spread	8	0.022 ± 0.002	4.7	0.007	0.497	0.857 ± 0.002	0.67	0.37	89.9
ensemble spread + HCU	8	0.020 ± 0.002	4.6	0.007	0.499	0.858 ± 0.002	0.67	0.37	89.7
post hoc comparators (added after the confirmatory analysis)									
latent distance $\bar{L}$	1	0.064 ± 0.004	12.1	0.026	0.406	0.789 ± 0.004	0.59	0.26	89.4
$\bar{L}$ + N_{atom}	1	0.035 ± 0.003	7.8	0.016	0.420	0.802 ± 0.004	0.62	0.27	89.8
HCU + $\bar{L}$	1	0.035 ± 0.003	7.3	0.014	0.425	0.807 ± 0.004	0.63	0.29	89.6

Table 8: Atlas-motif error ratios (motif / rest) on held-out nine-heavy-atom molecules for the size-extrapolating models and for random-split models trained with nine-heavy-atom molecules in their training distribution (means over seeds).

	≤8 heavy (OOD)	random 21,799	random 50,000	random 110,426
MAE (kcal mol^{-1})	0.731 ± 0.007	0.565 ± 0.008	0.360 ± 0.007	0.276 ± 0.009
failures >1 kcal mol^{-1} (%)	17.1	13.1	5.2	2.6
molecules	64,539	53,870	39,939	10,044
S13 fused bicyclic	1.97	1.49	1.42	1.35
S08 fused small ring	1.57	1.33	1.31	1.28
S04 4-membered ring	1.36	1.24	1.21	1.20
S11 cage (≥3 rings)	2.28	1.77	1.65	1.55
N04 novel Murcko scaffold	3.71	2.39	2.33	2.20
N03 novel ring system	4.44	2.71	2.62	2.44
S05 oxetane/azetidine	1.58	1.44	1.36	1.31
S09 bridged polycycle	2.23	1.67	1.54	1.48
G03 compressed 1,3 contact	2.19	1.79	1.62	1.50
N01 novel radius-1 environment	1.89	1.62	1.59	1.43
G05 sp^2 angle <100°	1.78	1.64	1.49	1.38
G04 stretched bond >1.58 Å	2.70	2.06	1.87	1.74
H15 ≥3 N atoms	1.60	1.60	1.42	1.33
H12 tertiary amine	1.44	1.30	1.32	1.34
S12 sp^2 ring-fusion atom	2.77	2.39	2.16	1.97
S10 spiro centre	2.21	1.52	1.41	1.40
S06 unsaturated 4-ring	1.50	1.44	1.33	1.28
E03 6-ring aromatic	1.24	1.30	1.14	1.06
H02 N-O bond	1.44	1.45	1.26	1.19
E14 amidine/guanidine	1.91	2.04	1.95	1.74
G06 non-planar amide N	1.94	2.01	1.96	1.87
E23 zwitterion	10.54	10.02	10.21	7.18
G02 short H$\cdots$H contact	1.59	1.49	1.54	1.52
E04 conjugated diene	1.68	1.62	1.48	1.42
G07 twisted conjugation	2.79	2.64	2.65	2.41

Table 9: Informativeness of R with and without the hierarchical regulariser, and in distribution (means ± s.e. over seeds). The first three rows are size-extrapolating models (seeds 0–2, identical splits) scored on the held-out evaluation molecules; random-split models are scored on their own test sets. AUROC_R, $\text{AUROC}_{\text{HCU}}$: area under the ROC curve for errors above 1 kcal mol^{-1} of R and of HCU refitted on discovery molecules for these models; last two columns: atlas motifs (frozen) with Holm-significant excess error and order-2 excess on held-out molecules. These re-analyses were added after the confirmatory analysis.

						atlas motifs significant	
Models	MAE (kcal mol^{-1})	mean R	$\rho(R, \lvert\varepsilon\rvert)$	AUROC_R	$\text{AUROC}_{\text{HCU}}$	error	$\Delta r^{(2)}$
$\lambda_R = 10^{-2}$ (main)	0.736 ± 0.016	0.16	0.324 ± 0.019	0.736 ± 0.011	0.767 ± 0.010	25/25	25/25
$\lambda_R = 0$	0.755 ± 0.016	3.03	0.044 ± 0.067	0.540 ± 0.048	0.713 ± 0.012	25/25	13/25
paper initialisation	0.713 ± 0.006	0.17	0.323 ± 0.007	0.745 ± 0.007	0.771 ± 0.006	25/25	25/25
random 21,799	0.545 ± 0.008	0.14	0.317 ± 0.002	0.763 ± 0.005	–	–	–
random 50,000	0.345 ± 0.008	0.12	0.284 ± 0.006	0.818 ± 0.008	–	–	–
random 110,426	0.262 ± 0.007	0.12	0.255 ± 0.005	0.848 ± 0.008	–	–	–

Table 10: rMD17 (PBE/def2-SVP, 500 K MD), all 100,000 conformations per molecule (azobenzene 99,988), eight models. Med. ΔE: median distortion energy above the lowest conformation; med. $|\tilde{\varepsilon}|$: median error of the relative conformational energy (kcal mol^{-1}); ρ: within-molecule Spearman correlations of R and of the HCU prediction μ with the error.

Molecule	heavy	atlas motifs	med. ΔE	med. $\lvert\tilde{\varepsilon}\rvert$	$\lvert\tilde{\varepsilon}\rvert/\Delta E$	R/N_{atom}	$\rho(R, \lvert\tilde{\varepsilon}\rvert)$	$\rho(\mu, \lvert\tilde{\varepsilon}\rvert)$
ethanol	3	none	10.5	7.1	0.71	0.120	-0.21	-0.16
malonaldehyde	5	N01	10.8	6.2	0.59	0.065	+0.11	+0.12
benzene	6	E03	7.7	4.1	0.54	0.058	+0.04	+0.05
toluene	7	E03	15.6	9.1	0.60	0.085	+0.15	+0.14
uracil	8	E03	13.8	11.2	0.83	0.041	+0.23	+0.17
naphthalene	10	E03, N03, N04, S13	21.5	21.4	1.00	0.122	-0.16	-0.20
salicylic	10	E03, N01	16.3	9.6	0.60	0.083	+0.13	+0.12
paracetamol	11	E03, N01	20.3	20.4	1.02	0.063	+0.19	+0.16
aspirin	13	E03, G07, N01	19.2	10.6	0.57	0.086	+0.05	+0.06
azobenzene	14	E03, N01, N04	20.6	10.8	0.54	0.080	+0.14	+0.09

Table 11: Conformational transfer of the atlas to ANI-1x (30,609 molecules, 3,112,861 conformations, eight models). p_{Holm}: stratified permutation test (4,000 permutations) of the distortion-adjusted outcome, Holm-adjusted over the 24 motifs.

		error ratio					ANI-1x ratio by distortion (kcal mol^{-1})				
Motif	mol.	QM9	ANI	raw	TR	p_{Holm}	<5	5–15	15–40	≥ 40	$\Delta r^{(2)} \times 10^3$
S04 4-membered ring	4574	1.22	1.01	1.10	0.06	0.294	1.05	1.00	1.02	1.07	+14
S05 oxetane/azetidine	3074	1.34	1.02	1.09	0.08	0.013	1.08	1.02	1.03	1.06	+27
S06 unsaturated 4-ring	704	1.27	0.85	0.85	-0.70	1.000	0.90	0.87	0.90	0.82	-18
E03 6-ring aromatic	1244	1.39	0.88	0.80	-0.37	1.000	0.92	0.96	0.97	0.77	-10
E04 conjugated diene	1425	1.60	0.97	0.97	-0.06	1.000	0.98	0.99	0.96	0.98	+10
E14 amidine/guanidine	6038	1.67	1.06	1.02	0.11	0.006	1.02	1.03	1.03	1.06	-8
H02 N-O bond	6779	1.44	1.07	0.97	0.17	0.006	0.96	1.02	1.07	0.99	+27
H12 tertiary amine	1265	1.31	1.04	1.18	0.14	0.007	1.19	1.01	1.04	1.15	+9
S08 fused small ring	7963	1.41	1.02	1.19	0.07	0.006	1.15	1.05	1.05	1.10	+2
S09 bridged polycycle	2431	1.72	1.07	1.23	0.13	0.006	1.09	1.07	1.11	1.16	+22
S10 spiro centre	621	1.50	0.96	1.06	-0.10	1.000	1.07	0.93	0.95	1.01	-36
S11 cage ($\geq$3 rings)	3267	1.83	1.05	1.24	0.08	0.006	1.13	1.07	1.10	1.15	+44
S12 sp^2 ring-fusion atom	1514	2.20	0.99	1.03	-0.01	1.000	0.96	1.00	1.01	0.99	+1
S13 fused bicyclic	8206	1.61	0.99	1.09	-0.03	1.000	1.12	1.03	1.02	1.00	-4
H15 $\geq$3 N atoms	10512	1.65	1.06	0.98	0.12	0.006	0.97	1.02	1.06	1.00	-1
G02 short H$\cdots$H contact	2277	1.64	1.11	1.27	0.22	0.006	1.14	1.13	1.10	1.23	+29
G03 compressed 1,3 contact	2498	1.70	1.02	1.07	0.04	0.013	1.07	1.02	1.05	1.04	+23
G04 stretched bond $>$1.58 Å	5460	1.79	1.11	1.25	0.18	0.006	1.21	1.13	1.12	1.18	+38
G05 sp^2 angle $<100^\circ$	5899	1.40	1.01	1.04	0.02	1.000	1.02	1.03	1.03	1.00	+5
G06 non-planar amide N	5294	1.68	1.04	1.04	0.07	0.006	1.03	1.03	1.02	1.05	+3
G07 twisted conjugation	1969	2.46	1.01	0.97	0.01	1.000	0.97	0.97	0.98	1.00	-9
N01 novel radius-1 environment	20065	1.52	1.04	0.97	0.09	0.006	0.93	0.99	1.01	1.01	+9
N03 novel ring system	13615	3.32	1.00	1.07	-0.00	1.000	1.09	1.06	1.04	0.99	+22
N04 novel Murcko scaffold	11878	2.72	0.98	0.99	-0.02	1.000	1.04	1.05	1.03	0.93	+10